



Sunbeam Institute of Information Technology

Pune and Karad

Module – Internet of Things (IoT)

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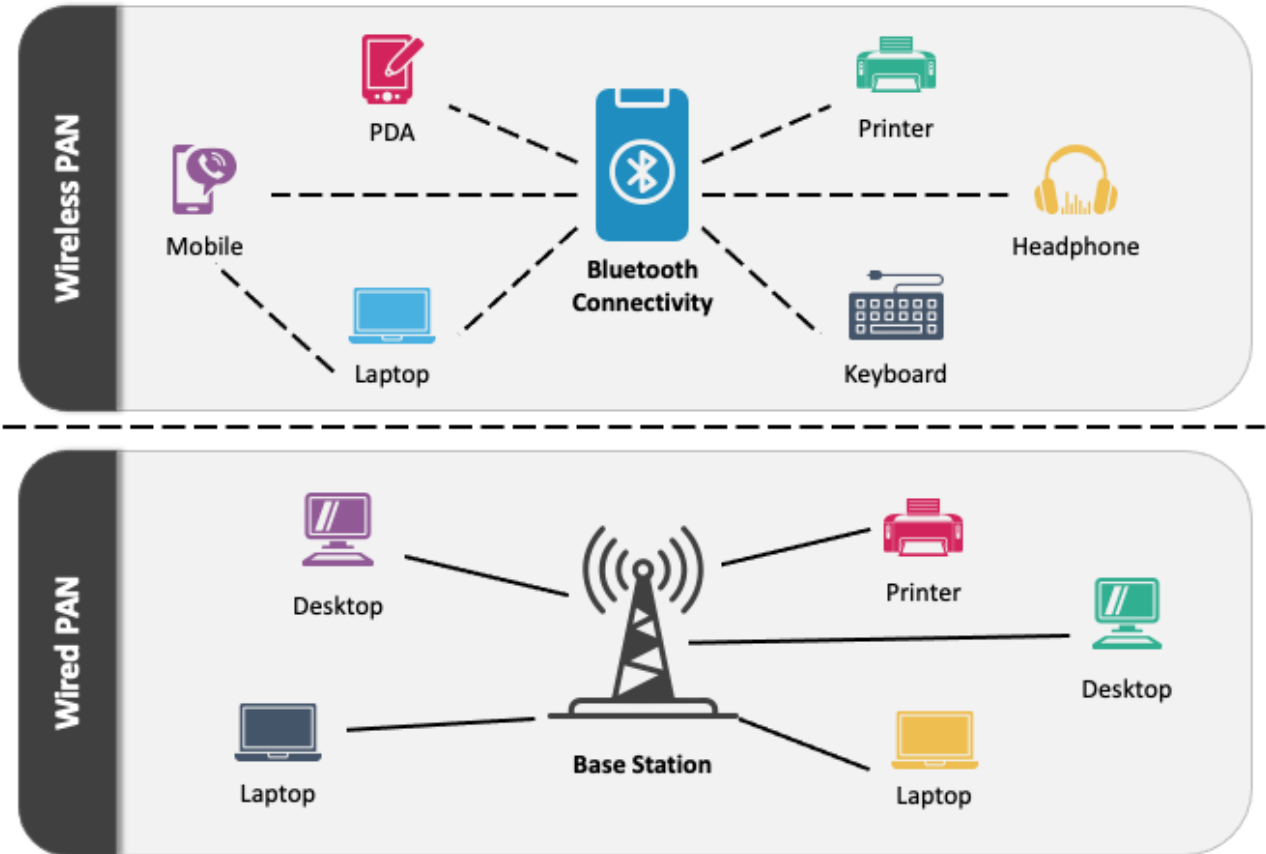
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Personal Area Network (PAN)

- A **personal area network (PAN)** is a computer network for interconnecting electronic devices within an individual person's workspace.
- A PAN provides data transmission among devices such as computers, smartphones, tablets and personal digital assistants.
- PANs can be used for communication among the personal devices themselves, or for connecting to a higher level network and the Internet where one master device takes up the role as gateway.

PERSONAL AREA NETWORK (PAN)

Types of PAN Network

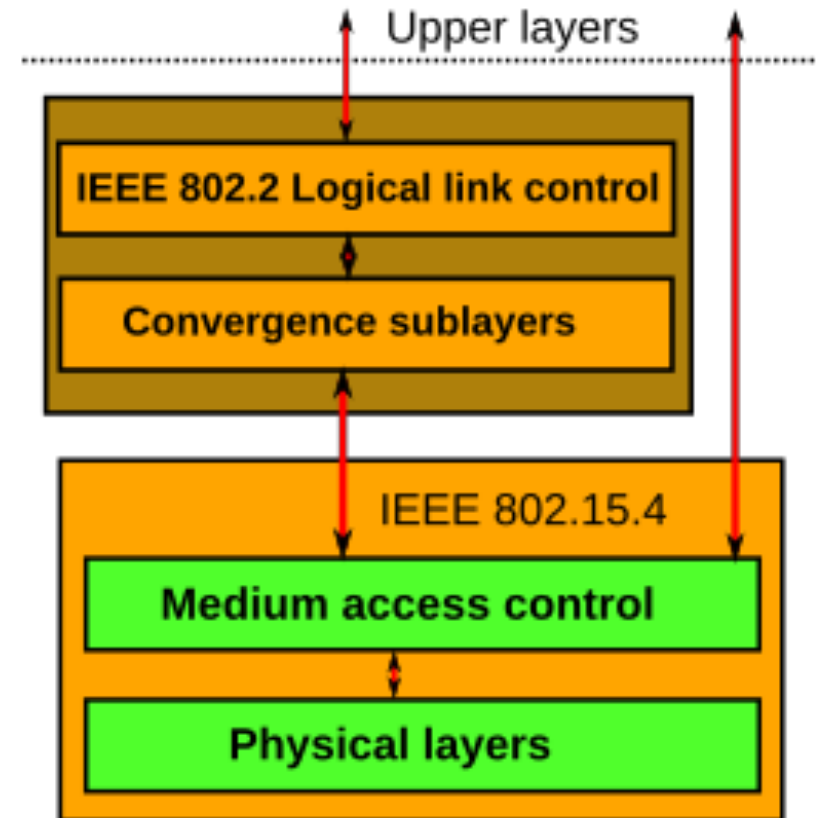


Personal Area Network (PAN)

- A PAN may be carried over wired interfaces such as USB, but is predominantly carried wirelessly, also called a **wireless personal area network (WPAN)**.
- A PAN is wirelessly carried over a low-powered, short-distance wireless network technology such as IrDA, Wireless USB, Bluetooth or Zigbee.
- A wireless personal area network (WPAN) is a personal area network in which the connections are wireless.
- IEEE 802.15 has produced standards for several types of PANs operating in the ISM band
- IEEE (Institute of Electrical and Electronics Engineers)
- specifies wireless personal area network (WPAN) standards.
 - IEEE 802.15.1: WPAN / Bluetooth
 - IEEE 802.15.2: Coexistence
 - IEEE 802.15.3: High Rate WPAN
 - IEEE 802.15.4: Low Rate WPAN
 - IEEE 802.15.5: Mesh Networking
 - IEEE 802.15.6: Body Area Networks
 - IEEE 802.15.7: Visible Light Communication
 - IEEE P802.15.8: Peer Aware Communications
 - IEEE P802.15.9: Key Management Protocol
 - IEEE P802.15.10: Layer 2 Routing
 - IEEE 802.15.13: Multi-Gigabit/s Optical Wireless Communications

- **IEEE 802.15.4** is a technical standard that defines the operation of a low-rate wireless personal area network (LR-WPAN).
- It specifies the physical layer and media access control for LR-WPANs, which defined the standard in 2003.
- IEEE standard 802.15.4 focuses on low-cost, low-speed communication between devices and even low power consumption. (contrast with other approaches, such as Wi-Fi, which offers more bandwidth and requires more power.)
- Key 802.15.4 features include:
 - Suitability for real-time applications with reservation of Guaranteed Time Slots (GTS)
 - Collision avoidance through CSMA/CA
 - Integrated support for secure communications
 - Power management functions
 - Support for time- and data-rate-sensitive applications
 - IEEE 802.15.4 devices may use one of three possible frequency bands for operation (868/915/2450 MHz).

- only the lower layers are defined in this standard.
- interaction with upper layers is done using an IEEE 802.2 logical link control sublayer accessing the MAC through a convergence sublayer.



IEEE 802.15.4 Protocol stack

- **Physical Layer**

- physical layer is the bottom layer in the OSI reference model
- The physical layer (PHY) provides the data transmission service.
- provides an interface to the physical layer management entity, which offers access to every physical layer management function maintains a database of information on related personal area networks.
- the PHY manages the physical radio transceiver, performs channel selection along with energy and signal management functions.
- It operates on one of three possible unlicensed frequency bands:
 - 868.0–868.6 MHz: Europe, allows one communication channel (2003, 2006, 2011)
 - 902–928 MHz: North America, originally allowed up to ten channels (2003), but since has been extended to thirty (2006)
 - 2400–2483.5 MHz: worldwide use, up to sixteen channels (2003, 2006)

- **MAC Layer**

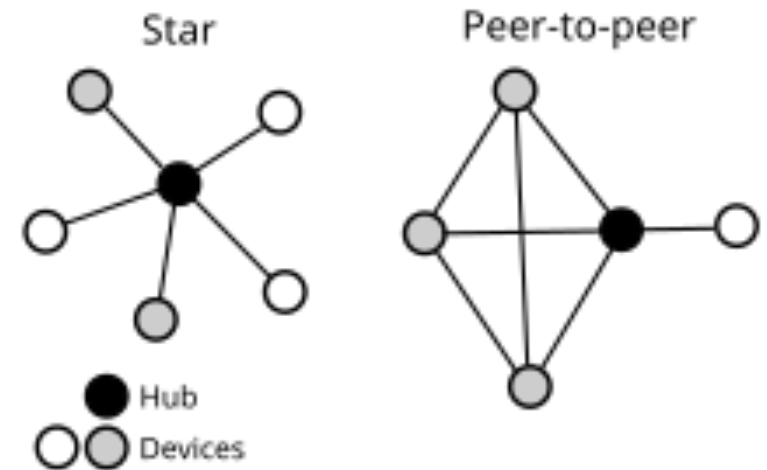
- enables the transmission of MAC frames through the use of the physical channel.
- it offers a management interface and itself manages access to the physical channel and network beacons.
- It also controls frame validation, guarantees time slots and handles node associations.

- **Node types**

- The standard defines two types of network node.
 - **full-function device** (FFD)
 - It can serve as the coordinator of a personal area network
 - It implements a general model of communication which allows it to talk to any other device
 - **reduced-function devices** (RFD)
 - extremely simple devices with very modest resource and communication requirements
 - they can only communicate with FFDs and can never act as coordinators.

- **Topologies**

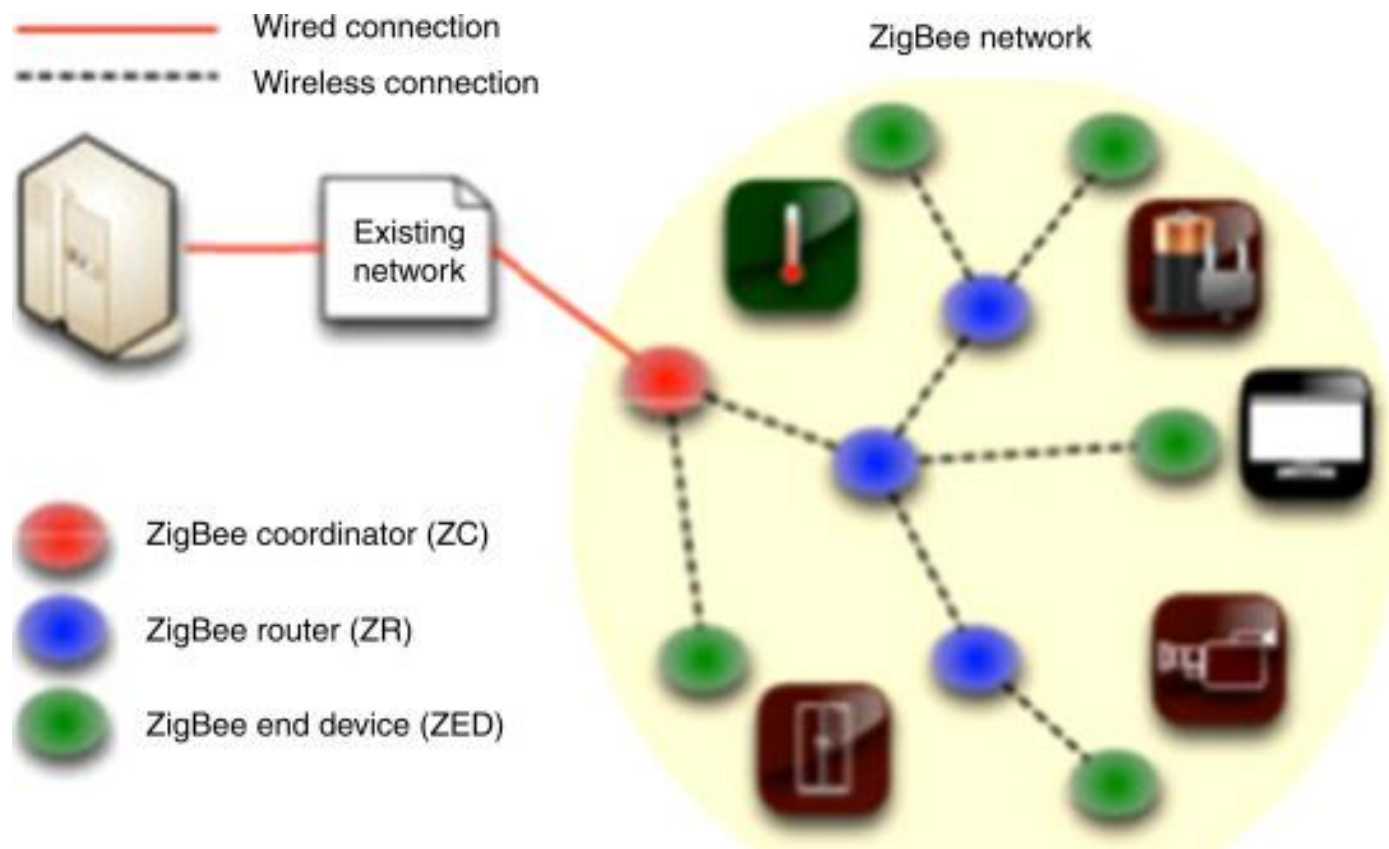
- Networks can be built as either peer-to-peer or star networks.
- However, every network needs at least one FFD to work as the coordinator of the network.
- Networks are thus formed by groups of devices separated by suitable distances.
- Each device has a unique 64-bit identifier, (sometimes, short 16-bit identifiers can be used within a restricted environment)



- Zigbee is an IEEE 802.15.4-based specification
- designed for small scale projects which needs wireless connection, low-power, low-bandwidth like low-power digital radios, home automation, medical device
- Hence, Zigbee is a low-power, low-data-rate, and close proximity (i.e., personal area)
- intended to be simpler and less expensive than other WPANs like Bluetooth, WiFi
- Zigbee was conceived in 1998, standardized in 2003, and revised in 2006.
- Zigbee is a low-power wireless mesh network standard targeted at battery-powered devices
- Zigbee operates in the industrial, scientific and medical (ISM) radio bands.
 - 2.4 GHz band being primarily used for lighting and home automation devices
- While devices for commercial utility use sub-GHz frequencies
 - 902-928 MHz in North America, Australia, and Israel
 - 868-870 MHz in Europe, 779-787 MHz in China
- data rates varying from around 20 kbit/s for sub-GHz bands to around 250 kbit/s for channels on the 2.4 GHz band range

Zigbee – Device Types

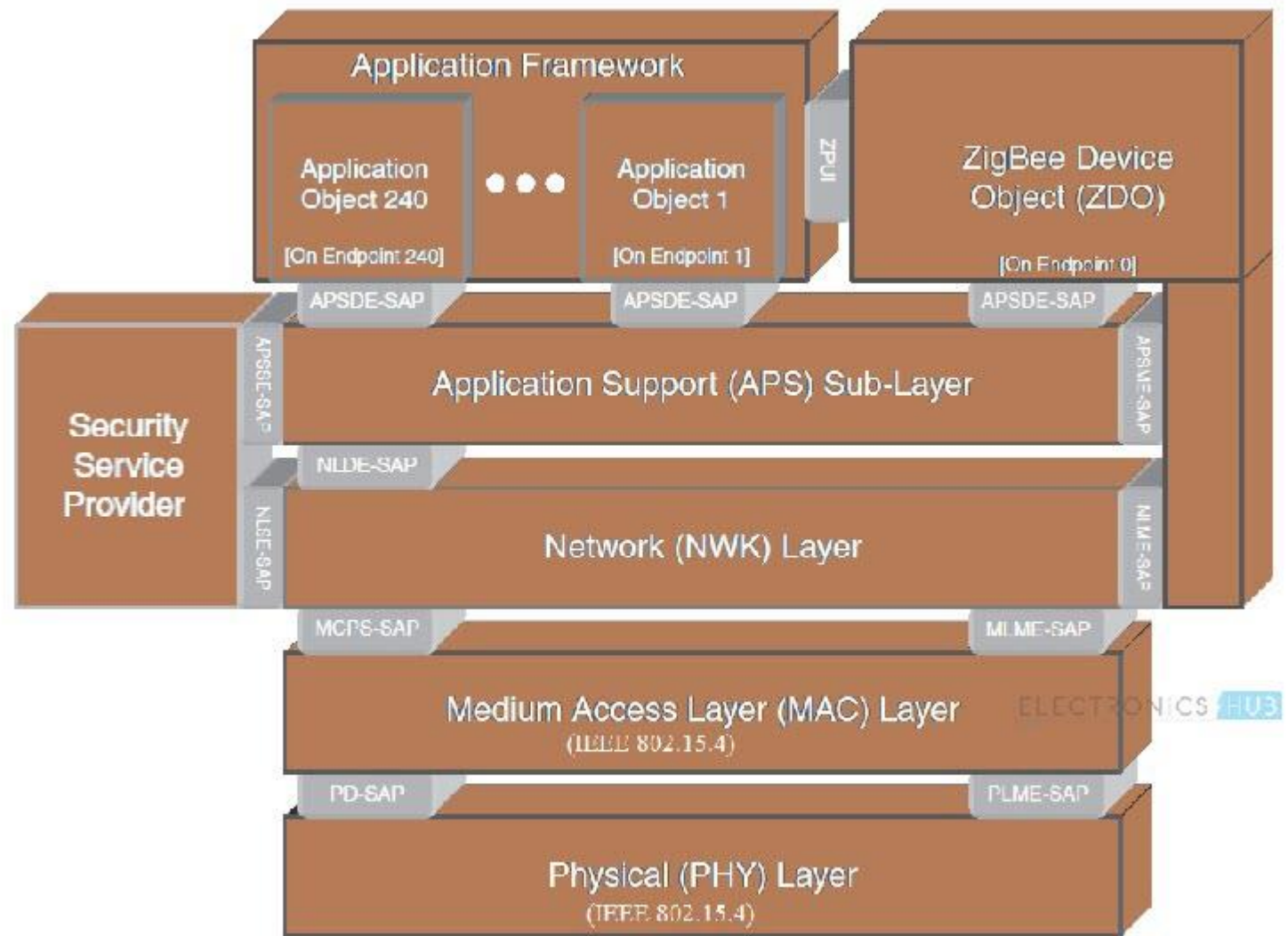
- **Zigbee coordinator (ZC)**
 - The most capable device, root of the network tree and may bridge to other networks.
 - There is precisely one Zigbee coordinator in each network
 - It stores information about the network, including acting as the trust center and repository for security keys.
- **Zigbee router (ZR)**
 - router devices can act as intermediate routers, passing data on to other devices.
 - These types of Zigbee products are typically mains-powered so they are always available on the network.
 - Zigbee Router devices are sometimes called Zigbee repeaters or Zigbee range extenders.
- **Zigbee end device (ZED)**
 - Contains just enough functionality to talk to the parent node (either the coordinator or a router)
 - it cannot relay data from other devices.
 - This relationship allows the node to be asleep a significant amount of the time thereby giving long battery life.
 - These types of Zigbee device products are often battery-powered.
 - A ZED requires the least amount of memory and thus can be less expensive to manufacture than a ZR or ZC.



- The Zigbee network layer natively supports both star and tree networks, and generic mesh networking.
- Every network must have one coordinator device.
- Within star networks, the coordinator must be the central node.
- Both trees and meshes allow the use of Zigbee routers to extend communication at the network level.

Zigbee – Protocol Architecture

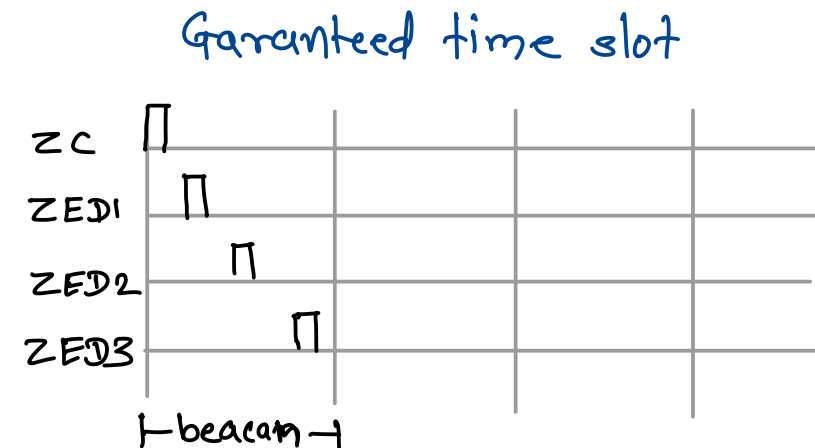
- Zigbee builds on the physical layer and media access control defined in IEEE standard 802.15.4
- The specification includes four additional key components
 - network layer
 - application layer
 - *Zigbee Device Objects* (ZDOs)
 - manufacturer-defined application objects



- **Network Layer**
 - [network layer](#) ensure correct use of the [MAC sublayer](#) and provide a suitable interface for upper layer
 - It deals with network functions such as connecting, disconnecting, and setting up networks.
 - It can establish a network, allocate addresses, and add and remove devices.
 - This layer makes use of star, mesh and tree topologies.
- **Application layer**
 - It is the highest-level layer defined by the specification and is the effective interface of the Zigbee system to its end users.
 - It comprises the majority of components added by the Zigbee specification
 - ZDO (Zigbee device object) and its management procedures
 - application objects defined by the manufacturer
 - This layer binds tables, sends messages between bound devices, manages group addresses, reassembles packets, and transports data.

- **ZDO (Zigbee device object)**
 - a protocol in the Zigbee protocol stack,
 - is responsible for
 - overall device management, security keys, and policies
 - defining the role of a device as either coordinator or end device
 - discovery of new devices on the network and the identification of their offered services
 - It may then go on to establish secure links with external devices and reply to binding requests accordingly.
- **Application Support Sublayer (APS)**
 - main standard component of the stack, and it offers a well-defined interface and control services.
 - It works as a bridge between the network layer and the other elements of the application layer
 - it keeps up-to-date binding tables in the form of a database (used to find appropriate devices depending on the services that are needed)
 - As the union between both specified layers, it also routes messages across the layers of the protocol stack.

- Typical application areas include:
 - Home automation
 - Wireless sensor networks
 - Industrial control systems
 - Embedded sensing
 - Medical data collection
 - Smoke and intruder warning
 - Building automation
 - Remote wireless microphone configuration



Non Guaranteed time slot
CSMA/CA

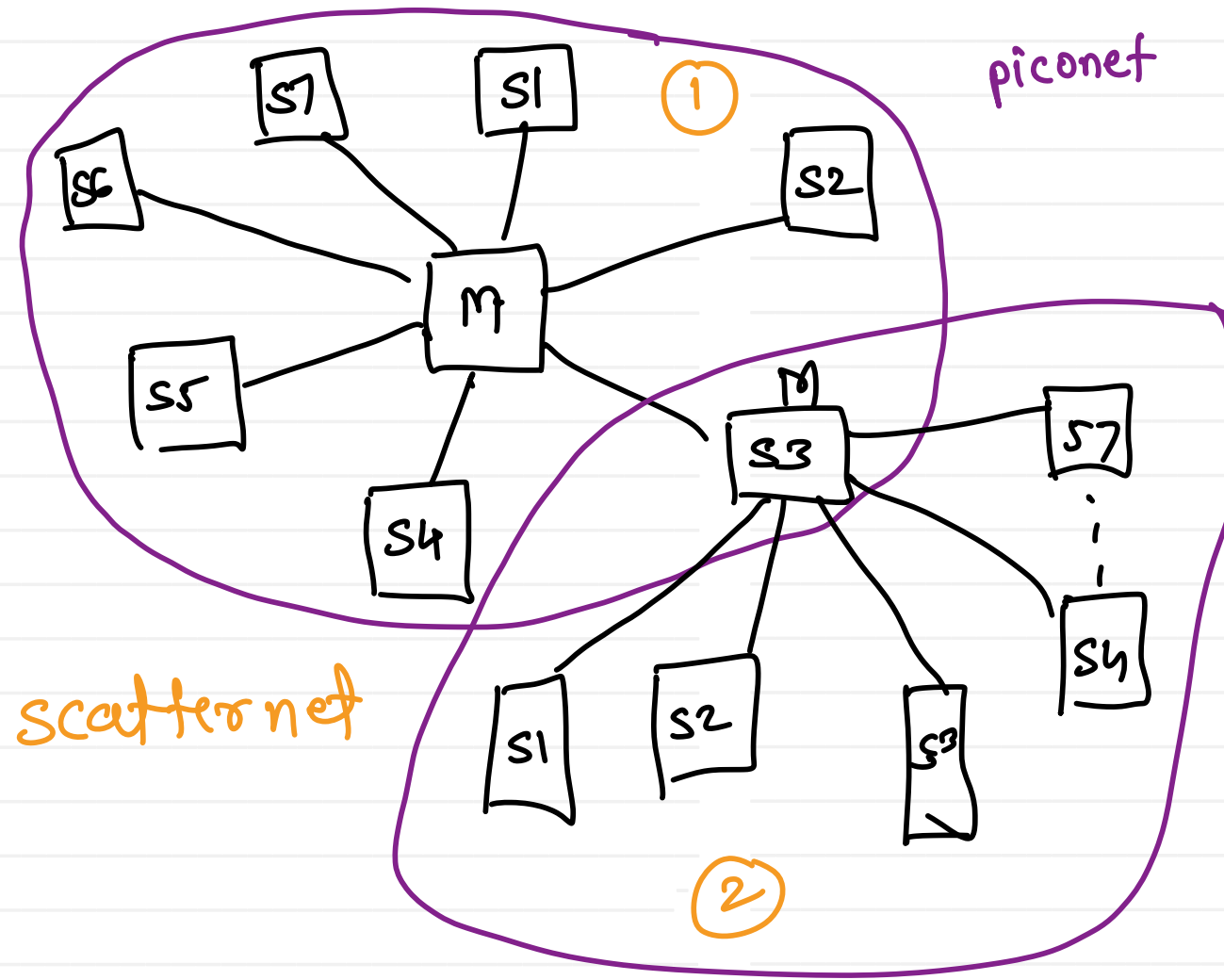
- Bluetooth technology is a high-speed low powered wireless technology
- It is a specification (IEEE 802.15.1) for the use of low-power radio communications
- Wireless signals transmitted with Bluetooth cover short distances, typically up to 30 feet (10 meters).
- Bluetooth wireless technology was named after a Danish Viking and King, Harald Blatand; his last name means "Bluetooth" in English. (He is credited with uniting Denmark and Norway)
- Emerged from the task undertaken by Ericsson Mobile Communications in 1994 to find an alternative to the use of cables for communication between mobile phones and other devices.
- In 1998, the companies Ericsson, IBM, Nokia, and Toshiba formed the Bluetooth Special Interest Group (SIG) which published the 1st version in 1999.

How Bluetooth works?

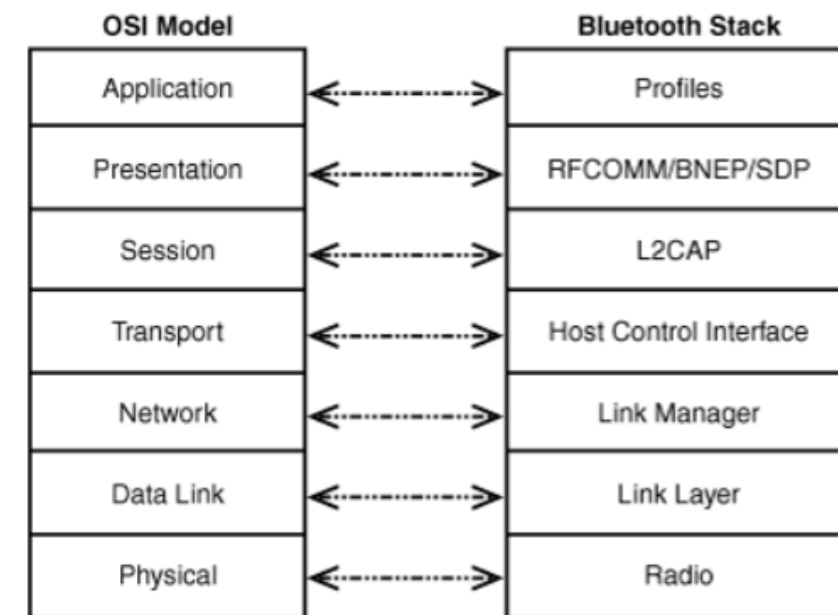
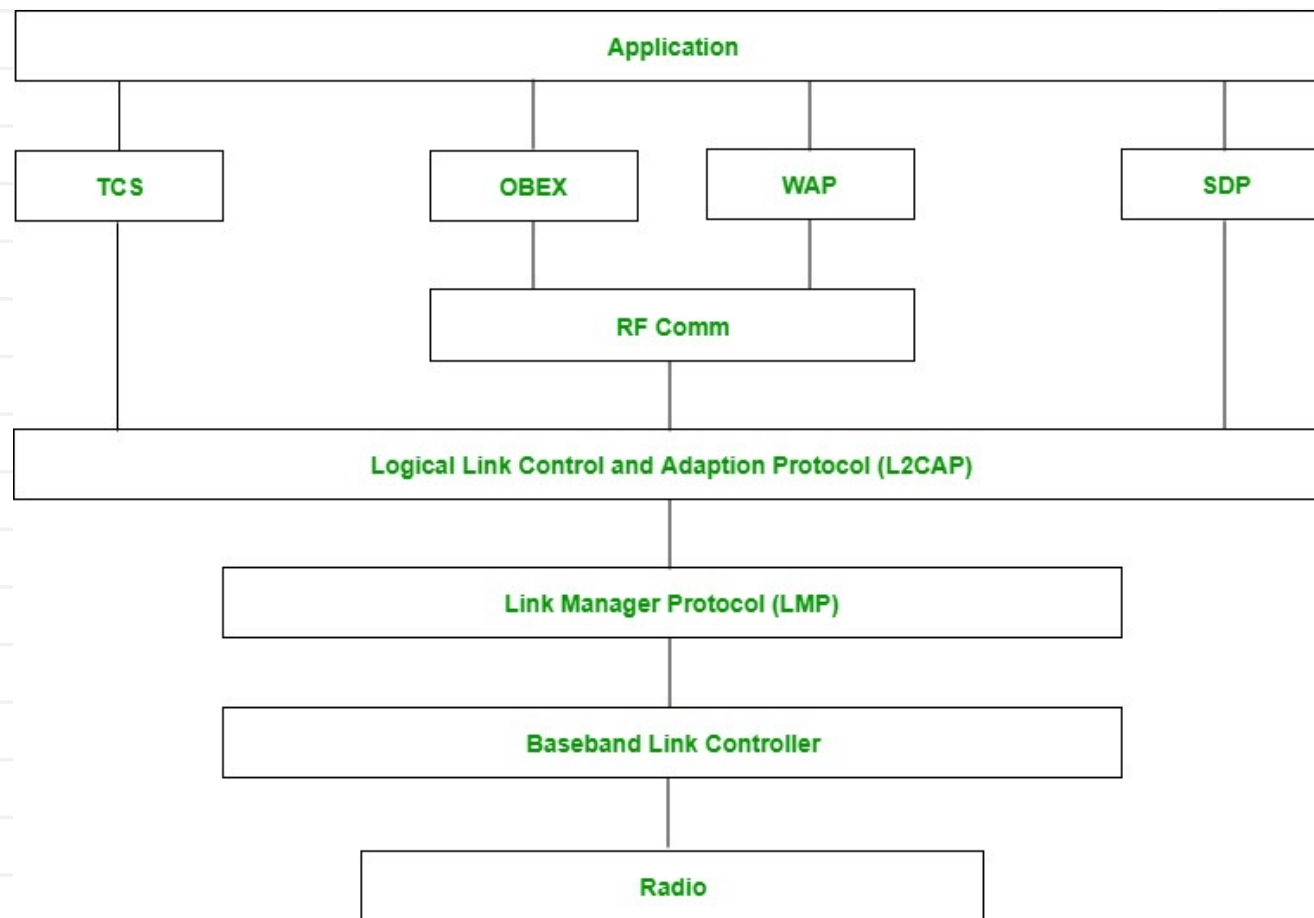
- Bluetooth Network consists of a minimum of 2 to a maximum of 8 Bluetooth peer devices. Usually a single master and up to 7 slaves.
- A master is a device that initiates communication with other devices.
- The master device governs the communications link and traffic between itself and the slave devices associated with it.
- A slave device is a device that responds to the master device.
- Transmissions by slave devices are governed by the master device (i.e., the master device dictates when a slave device may transmit).
- A slave may only begin its transmissions in a time slot immediately following the time slot in which it was addressed by the master, or in a time slot explicitly reserved for use by the slave device.
- The frequency hopping sequence is defined by the Bluetooth device address (BD_ADDR) of the master device.
- The master device first sends a radio signal asking for a response from the particular slave devices within the range of addresses.
- The slaves respond and synchronize their hop frequency as well as a clock with that of the master device.

- The Bluetooth architecture uses two networks like Piconet and Scatternet
- **Piconet Network**
 - Piconet includes one main node (master node) and seven secondary nodes (slave nodes).
 - There are eight active nodes totally which are arranged at a 10 meters distance.
 - The message between these two nodes can be done one-to-one otherwise one-to-many.
 - Communication can be possible only among the master and slave but the communication like Slave-slave cannot be possible.
 - It also includes 255 parked nodes which are known as secondary nodes. These cannot communicate until it gets altered to the active condition.
- **Scatternet Network**
 - The formation of the Scatternet Network can be done through various piconets.
 - Slave in one piconet acts as a master in another piconet.
 - This type of node gets a message from the master within one piconet & transmits the message toward its slave in another piconet.
 - This kind of node is called a bridge-node. In two piconets, a station cannot be master.

Bluetooth



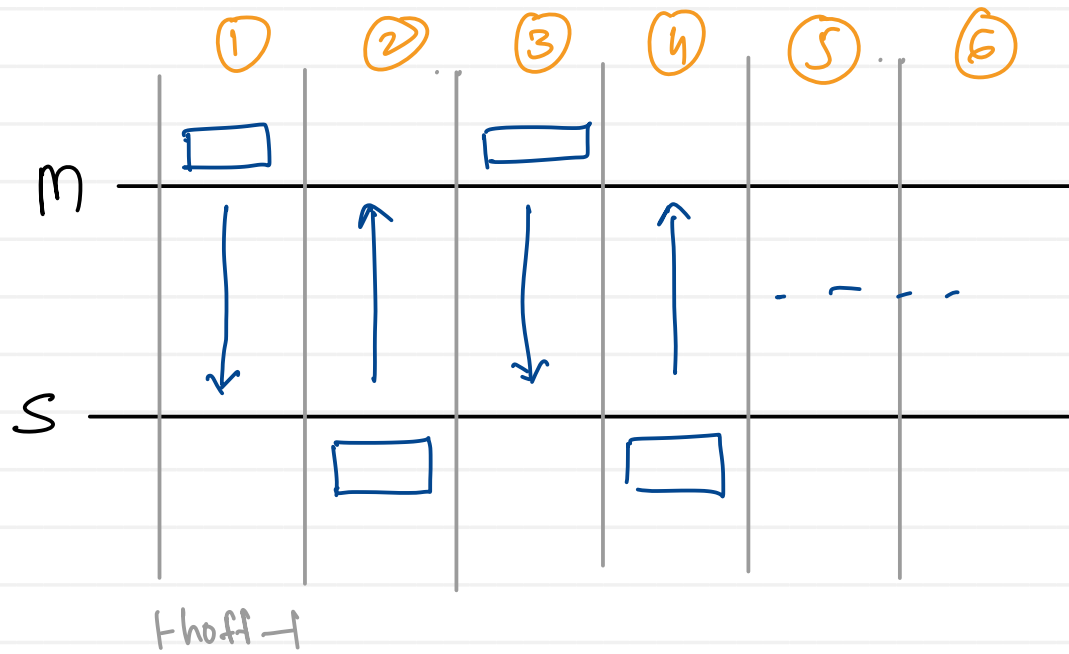
- The **Bluetooth specifications** include the following.
- **Core Specifications:**
 - It defines the Bluetooth protocol stack and the requirements for testing and qualification of Bluetooth-based products.
- **The Profiles Specification:**
 - It defines usage models that provide detailed information about how to use the Bluetooth protocol for various types of applications.



Bluetooth Core Specifications

- The core specifications of Bluetooth consists of 5 layers
- **Radio:** Radio specifies the requirements for radio transmission – including frequency, modulation, and power characteristics – for a Bluetooth transceiver.
- **Baseband Layer:** It defines physical and logical channels and link types (voice or data); specifies various packet formats, transmit and receive timing, channel control, and the mechanism for frequency hopping (hop selection) and device addressing. It specifies point to point or point to multipoint links. The length of a packet can range from 68 bits (shortened access code) to a maximum of 3071 bits.
- **LMP- Link Manager Protocol (LMP):** It defines the procedures for link setup and ongoing link management.
- **Logical Link Control and Adaptation Protocol (L2CAP):** It is responsible for adapting upper-layer protocols to the baseband layer.
- **Service Discovery Protocol (SDP):** – Allows a Bluetooth device to query other Bluetooth devices for device information, services provided, and the characteristics of those services.

Bluetooth



Advantages of Bluetooth

Spread

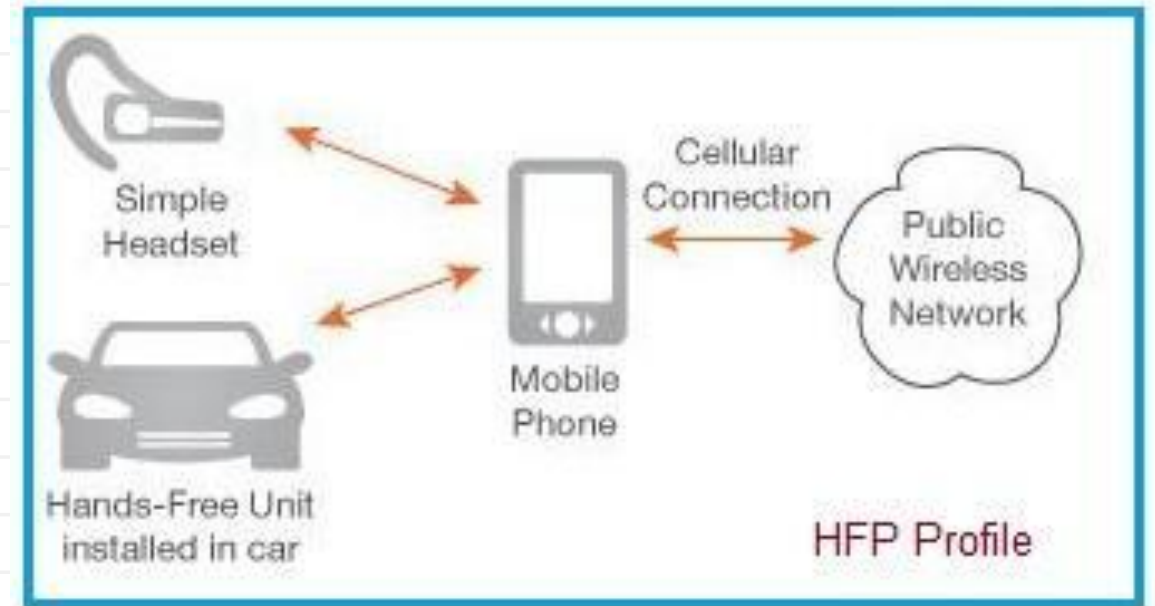
- It removes the problem of radio interference by using a technique called Spread Frequency Hopping. This technique utilizes 79 channels of the particular frequency band, with each device accessing the channel for only 625 microseconds, i.e. the device must toggle between transmitting and receiving data from one-time slot to another.
- This implies the transmitters change frequencies 1,600 times every second, meaning that more devices can make full use of a limited slice of the radio spectrum. This ensures that the interference won't take place as each transmitter will be on different frequencies.
- The power consumption of the chip (consisting of a transceiver) is low, at about 0.3mW, which makes it possible for the least utilization of battery life.
- It is possible to use Bluetooth for both transferring data and verbal communication as Bluetooth can support data channels of up to 3 similar voice channels.
- It overcomes the constraints of the line of sight and one-to-one communication as in another mode of wireless communications like infrared.

BTLE (Bluetooth Low Energy)

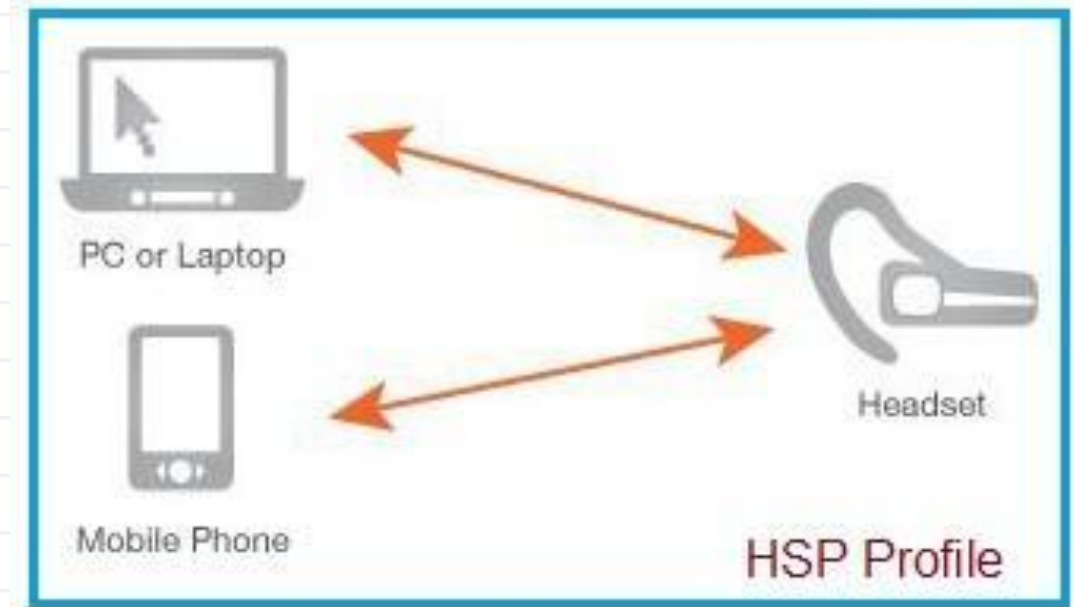
- a wireless personal area network technology designed and marketed by the Bluetooth Special Interest Group (Bluetooth SIG)
- Aimed at applications in the healthcare, fitness, beacons, security, and home entertainment industries.
- Compared to Classic Bluetooth, Bluetooth Low Energy is intended to provide considerably reduced power consumption and cost while maintaining a similar communication range.
- The original specification was developed by Nokia in 2006 under the name Wibree, which was integrated into Bluetooth 4.0 in December 2009 as Bluetooth Low Energy.
- Mobile operating systems including iOS, Android, Windows Phone and BlackBerry, as well as macOS, Linux, Windows 8, Windows 10 and Windows 11, natively support Bluetooth Low Energy.

- Bluetooth profiles serve various communication and entertainment needs.
- They ensure compatibility and smooth interactions between different Bluetooth-enabled devices.
- The choice of profile depends on the intended use case, such as
 - making calls
 - listening to music
 - controlling media playback
 - accessing contact and message data.

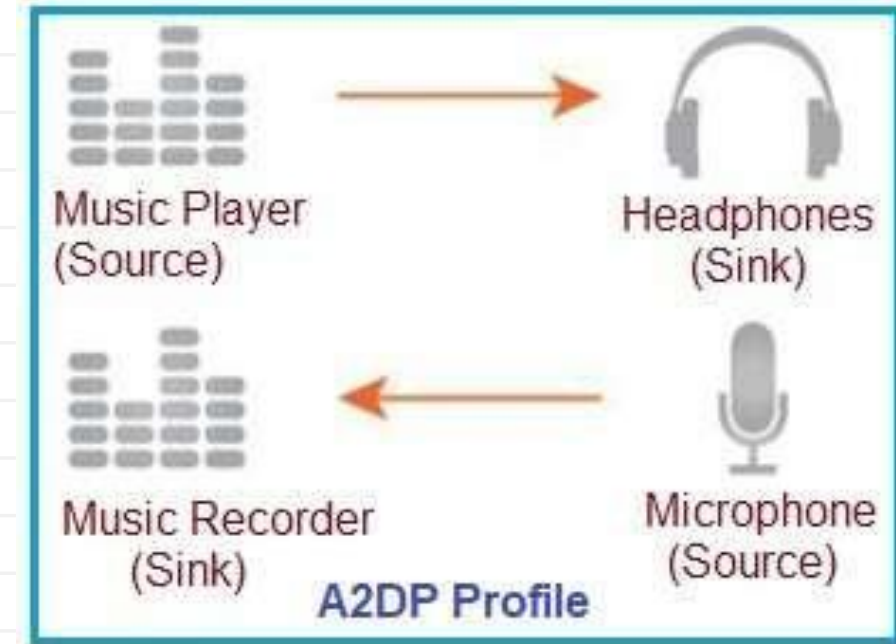
- HFP is designed for hands-free calling in vehicles and headsets.
- It allows for voice calls and audio streaming. It provides call control functions such as answer, end, and reject calls
- HFP stands for Hands-Free Profile
- Audio is transmitted over a synchronous SCO channel. A separate data channel is used to control the audio stream.
- Other features include support for mono audio, sampling rates from 8 KHz to 16 KHz, and an audio delay of about 20 to 30 ms.
- The HFP Bluetooth profile supports audio codecs such as CVSD and mSBC.
- AT commands are used to control the mobile phone.



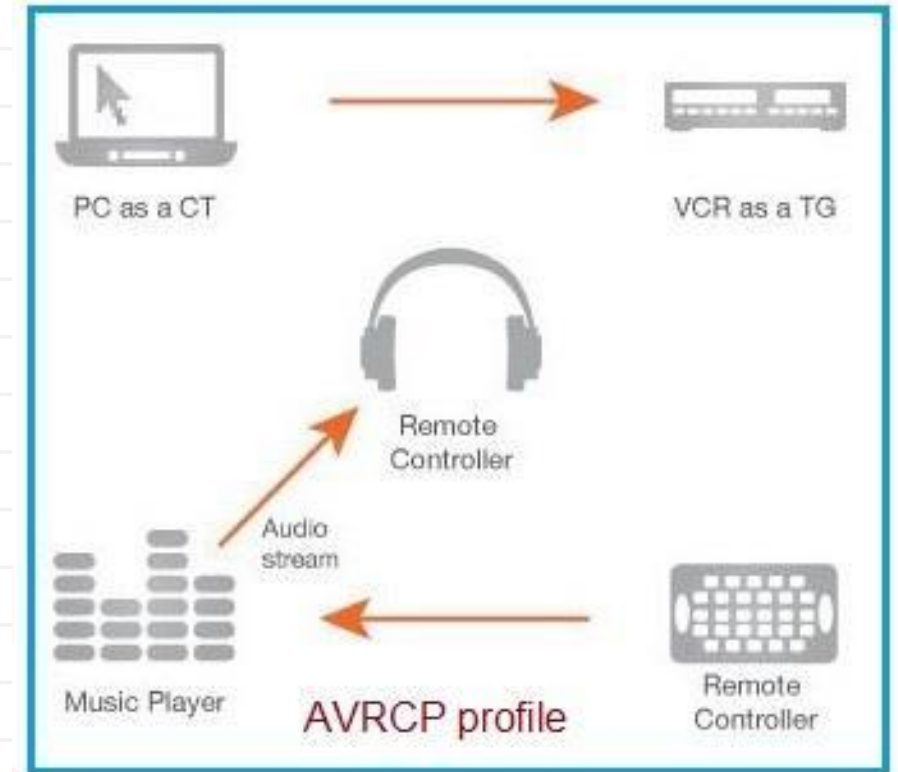
- It enables voice transmission between a mobile phone and a wireless Bluetooth headset.
- The primary use case is mono voice calls using Bluetooth headsets.
- It supports limited audio streaming capabilities and is often used in mono Bluetooth headsets.
- HSP stands for Headset Profile.
- Audio is transmitted over a synchronous SCO channel.
- It has become obsolete due to the availability of the HFP profile.
- Other features of the HSP Bluetooth profile are mono audio, 8 KHz sampling, audio delay of 20 to 30 ms, and CVSD codec, etc.



- A2DP stands for Advanced Audio Distribution Profile.
- It enables stereo audio streaming between a music player (as source) and speakers/headphones (as sink).
- It provides high-quality audio streaming (stereo). It is ideal for music playback from devices to Bluetooth audio output devices.



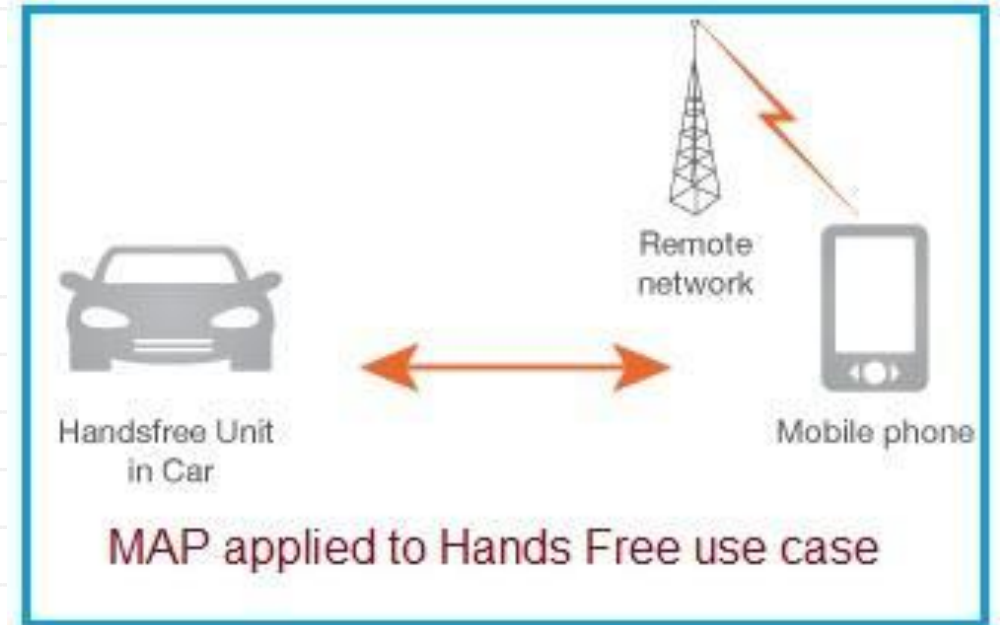
- AVRCP stands for Audio/Video Remote Control Profile.
- It enables an audio sink to control the audio sources' music player and streaming status.
- It uses a data channel to transmit control information between the audio controller and target.
- AVRCP v 1.5 enables advanced features like media browsing, audio player, and playlist management.
- Other features of the AVRCP Bluetooth profile are as follows:
 - Content browsing (artists, songs, etc.)
 - Content searching
 - Audio player management
 - Enables audio stream status control
 - Enables transmission of track, title, and other media information
 - Playlist management, etc.



- PBAP provides access to contact and address book information stored on a mobile device, typically used in car infotainment systems for accessing phone contacts and caller ID information.
- PBAP stands for Phone Book Access Profile.
- This profile enables the exchange of phone book objects between, for example, a smartphone and a car kit.
- It uses a data channel to transmit vCARDS over a Bluetooth connection.
- The other features of the PBAP Bluetooth profile are:
 - Download phone book items
 - Access call history
 - Access subscriber number information, etc



- MAP stands for Message Access Profile.
- MAP allows access to text messages and email data on a paired device, often used in car infotainment systems for hands-free access to messages and emails.
- The primary use case is accessing text messages and email data while driving through the car's infotainment system.
- This profile enables the exchange of messages between a smartphone and a car kit.
- It uses a data channel to transmit SMS and email notifications and messages over Bluetooth.
- The other features of the MAP Bluetooth profile are as follows:
 - SMS and email notifications
 - Download of messages
 - Upload of messages
 - Browsing message folders
- Bluetooth MAP Version specifications are developed to allow the exchange of messages between devices. It is mostly used for automotive hands-free use.

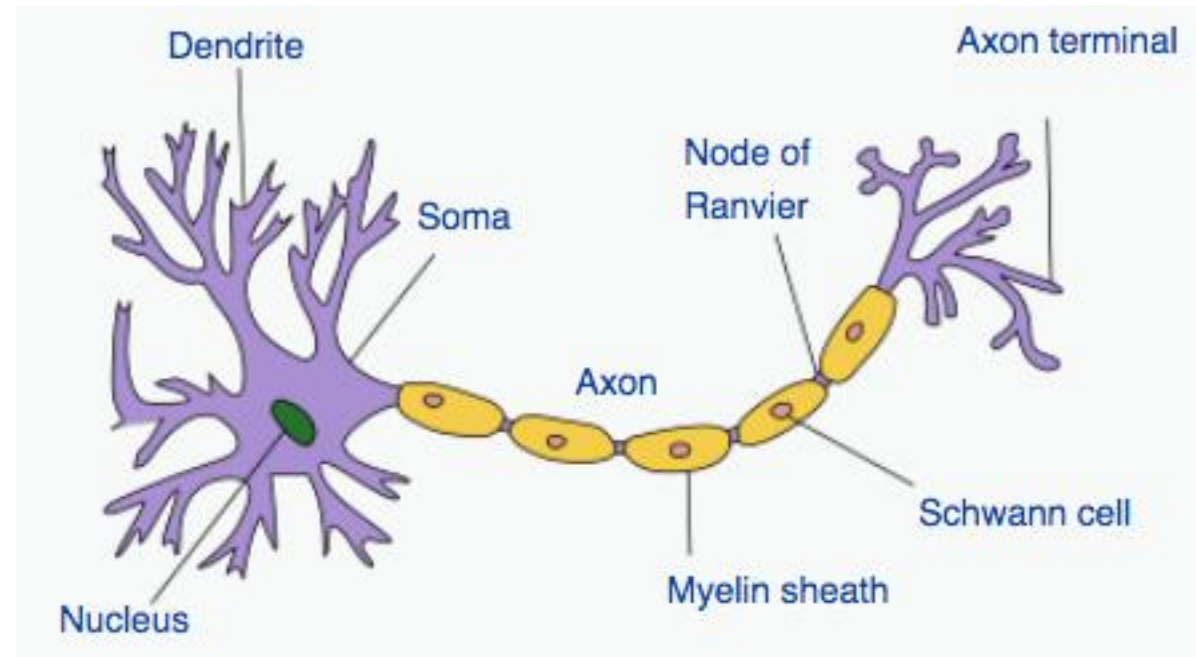


Profile	Full Form	Description
HFP	Hands-Free Profile	Allows for hands-free calling in cars and headsets.
HSP	Headset Profile	Basic audio communication between headsets and phones.
A2DP	Advanced Audio Distribution Profile	High-quality audio streaming between devices.
AVRCP	Audio/Video Remote Control Profile	Allows remote control of media playback.
PBAP	Phone Book Access Profile	Access to contact and address book information.
MAP	Message Access Profile	Access to text messages and email.

- 1) Convolution Neural Network (CNN)
 - images, audio & video
- 2) Recurrent Neural Network (RNN)
 - text processing (classification)
- 3) Artificial Neural Network (ANN)
 - numeric

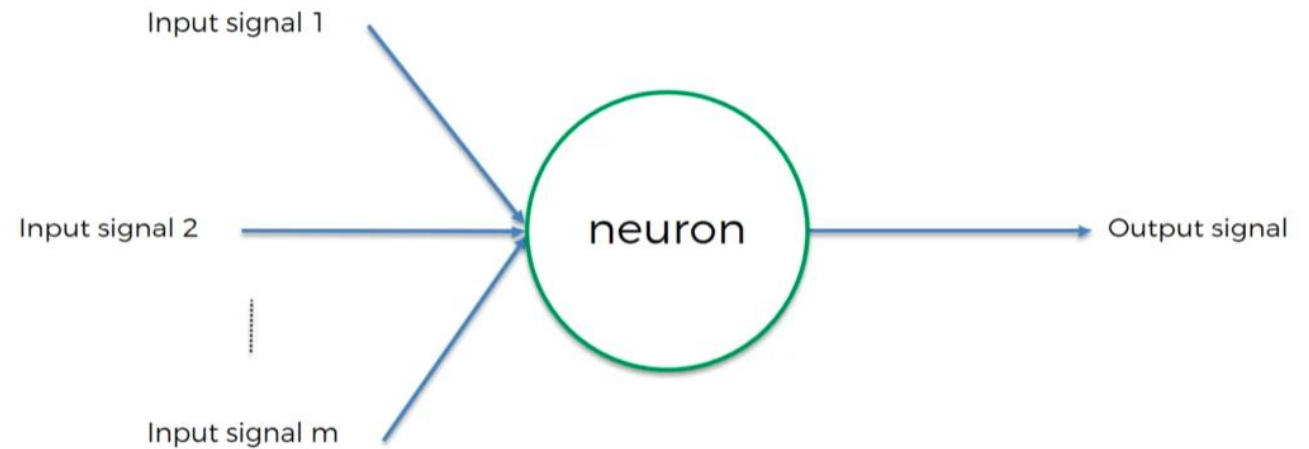
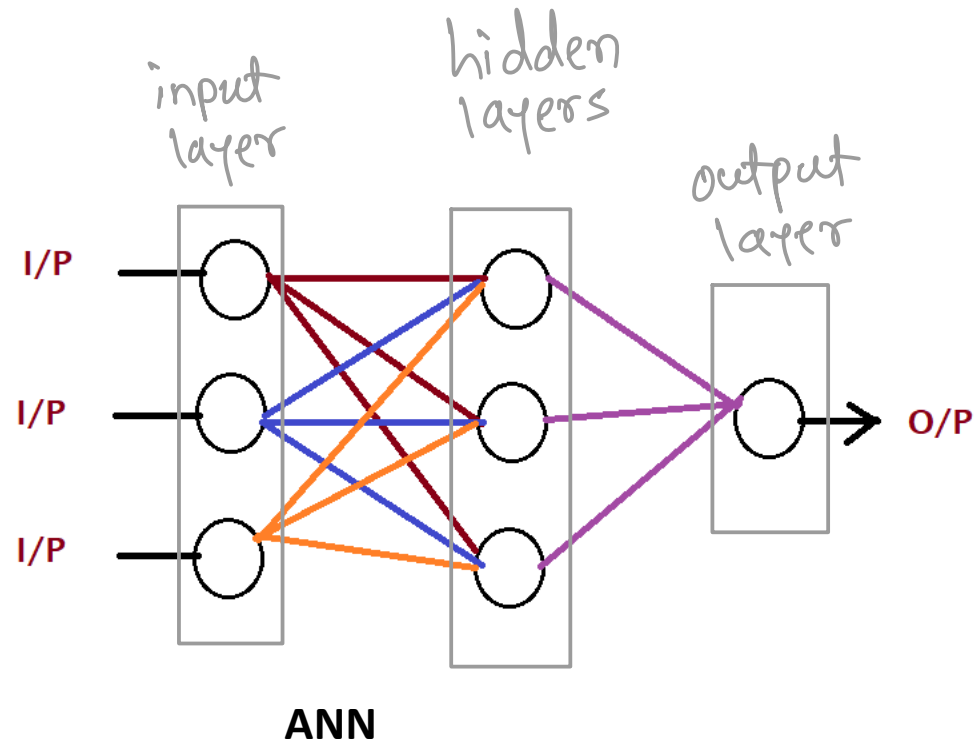
Neuron

- A neuron or nerve cell, is an electrically excitable cell that communicates with other cells via specialized connections called synapses
- A typical neuron consists of a cell body (soma), dendrites, and a single axon
- The soma is the body of the neuron
- The dendrites of a neuron are cellular extensions with many branches
- The axon primarily carries nerve signals away from the soma, and carries some types of information back to it



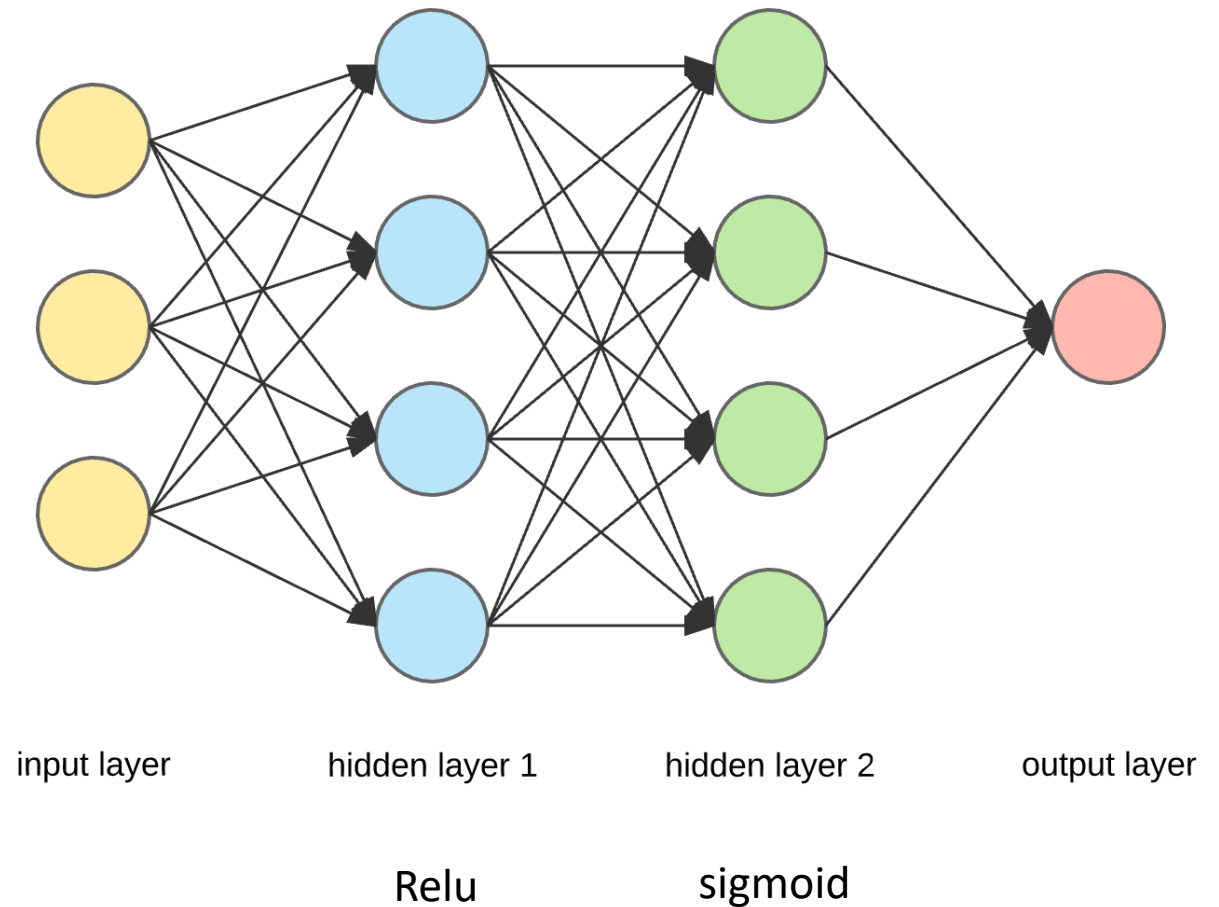
Artificial Neural Network

- Artificial neural networks (ANN) or connectionist systems are computing systems vaguely inspired by the biological neural networks that constitute animal brains
- The basic building block is a neuron



Artificial Neural Network

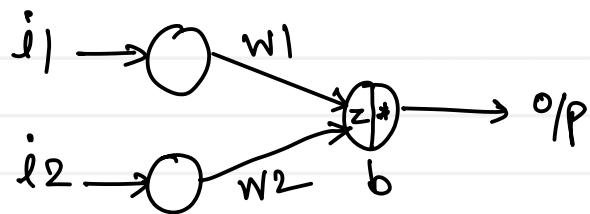
- Artificial Neural Networks (ANN) are multi-layer fully-connected neural nets
- They consist of an input layer, multiple hidden layers, and an output layer
- Every node in one layer is connected to every other node in the next layer
- We can make the network deeper by increasing the number of hidden layers.



Characteristics of Artificial Neural Network

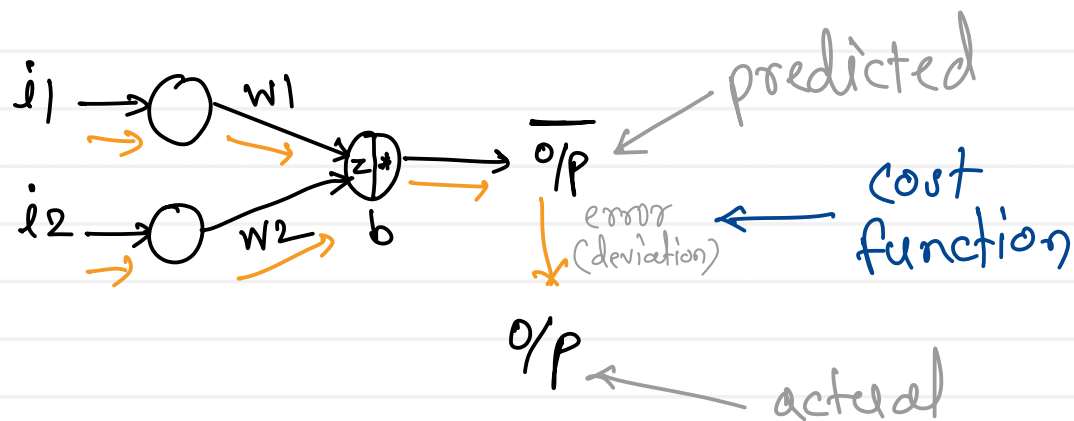
- It is neurally implemented mathematical model
- It contains huge number of interconnected processing elements called neurons to do all operations
- Information stored in the neurons are basically the weighted linkage of neurons
- The input signals arrive at the processing elements through connections and connecting weights.
- It has the ability to learn, recall and generalize from the given data by suitable assignment and adjustment of weights.
- The collective behavior of the neurons describes its computational power, and no single neuron carries specific information



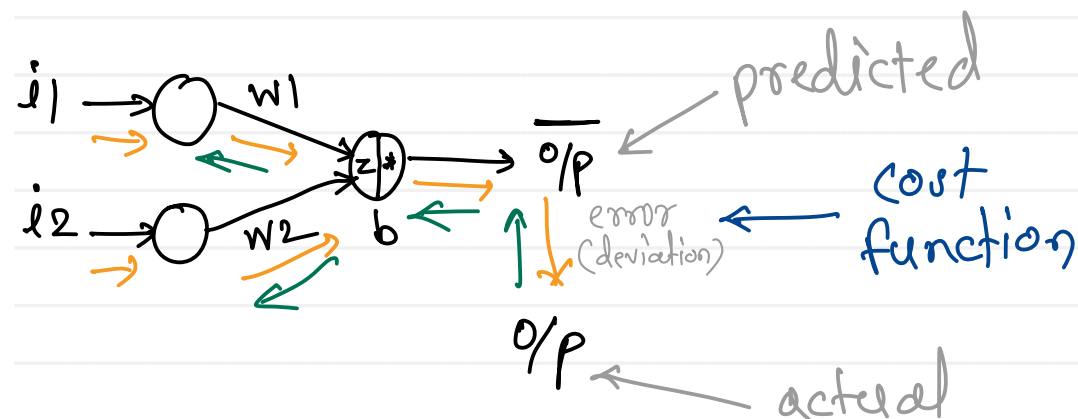


$$z = i_1 w_1 + i_2 w_2 + b$$

- 1) Threshold function
 - 2) ReLU (Rectified Linear Unit) function ← hidden layers
 - 3) Sigmoid function
 - 4) Soft max function
 - 5) Hyperbolic tangent (Tanh) functions
- } ← output layer



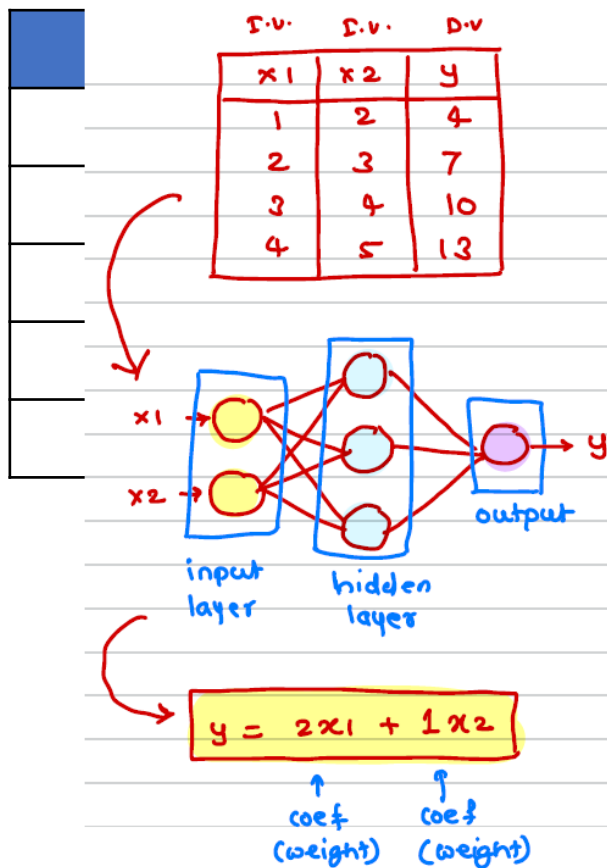
forward propagation



backward propagation

$$y = xw + b$$

$y = 0 \quad (2)$
 $y = 1 \quad (2)$
 $y = 2 \quad (2)$



$w = 0, 1, 2$

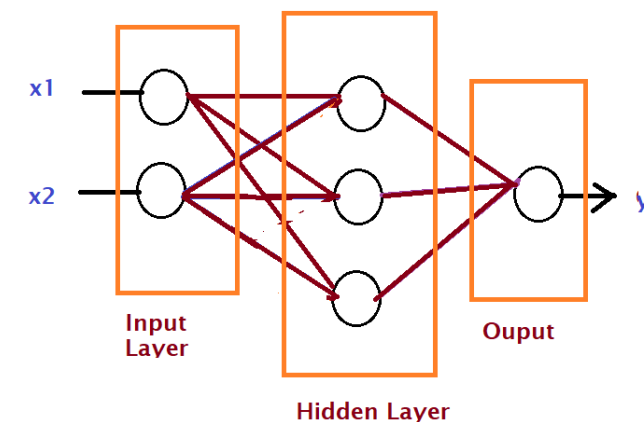
X	Y
1	2
2	4
3	6
4	8
16	?

$$Y = 2x$$

$$Y = 2 * x$$

Coefficient
(weight)

X1 (input variable)	X2 (input variable)	Y (dependent variable)
1	2	4
2	3	7
3	4	10
4	5	13



$$Y = 2 * x_1 + 1 * x_2$$

Coefficient
(weight)

Coefficient
(weight)

Trial (EPOCH) : 1

$$Y = 1 * x_1 + 1 * x_2$$

$$y = 1 + 2 = 3$$

$$y = 3 + 7 = 10$$

Trial (EPOCH) :2

$$Y = 2 * x_1 + 1 * x_2$$

$$y = 2 * 1 + 1 * 2 = 4$$

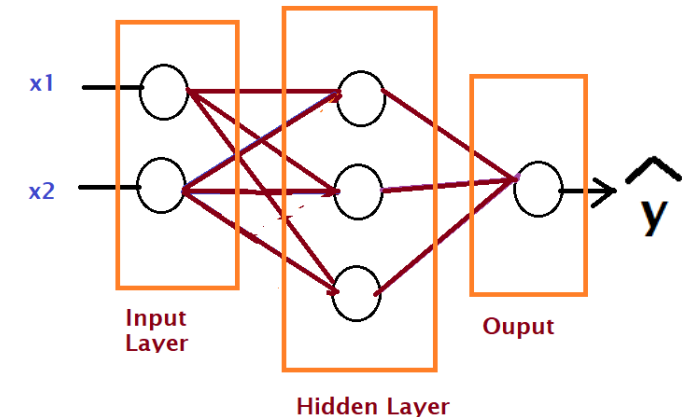
$$y = 2 * 2 + 1 * 3 = 7$$

It proves the formula

$Y = 2 * x_1 + 1 * x_2$ is giving correct prediction

Output / predicted value is represented as $\hat{y}$

X1 (input variable)	X2 (input variable)	Y (dependent variable)
1	2	4
2	3	7
3	4	10
4	5	13



$$Y = 2 * x_1 + 1 * x_2$$

Coefficient
(weight)

Coefficient
(weight)



Application of Neural Network

- Neural network is suitable for the research on *Animal behavior, predator/prey relationships etc.*
- It would be easier to do *proper valuation* of property, buildings, automobiles, machinery etc. with the help of neural network.
- Neural Network can be used in betting on horse races, sporting events and most importantly in stock market .
- It can be used to predict the correct judgement for any crime by using a large data of crime details as input and the resulting sentences as output.
- By analyzing data and determining which of the data has any fault (files diverging from peers) called as *Data mining, cleaning and validation* can be achieved through neural network.
- Neural Network can be used to predict targets with the help of echo patterns we get from sonar, radar, seismic and magnetic instruments .
- It can be used efficiently in *Employee hiring* so that any company can hire right employee depending upon the skills the employee has and what should be it's productivity in future .
- It has a large application in *Medical Research* .
- It can be used to for *Fraud Detection* regarding credit cards , insurance or taxes by analyzing the past records

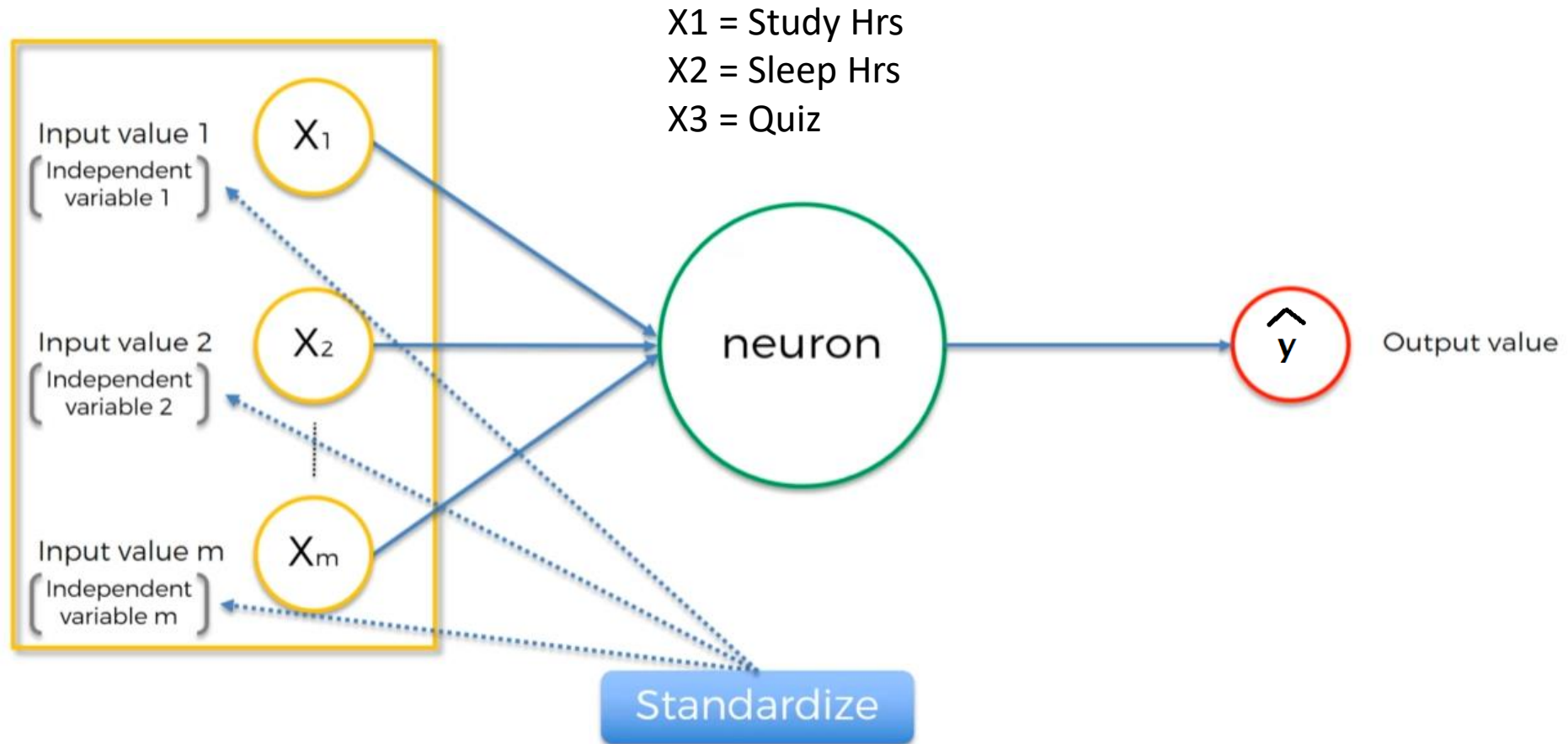


How does it work?

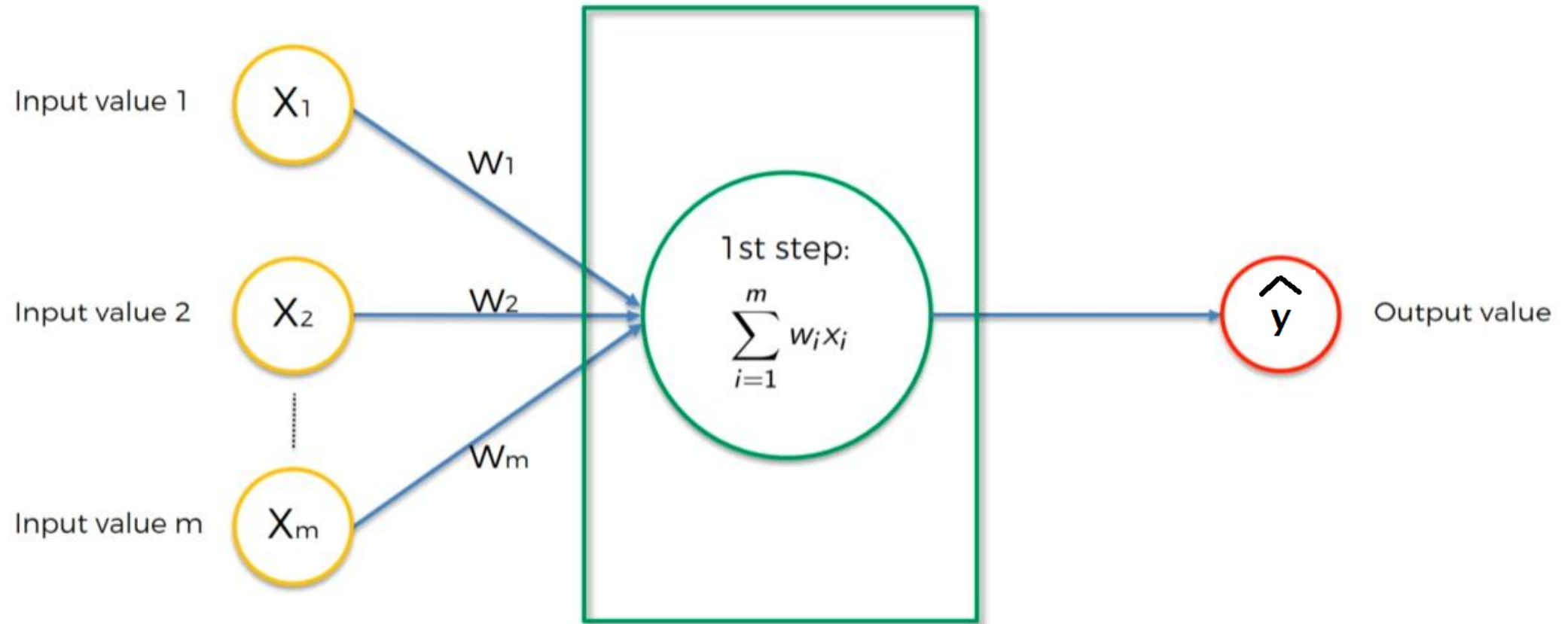
Row ID	Study Hrs	Sleep Hrs	Quiz	Exam
1	12	6	78%	93%
2	22	6.5	24%	68%
3	115	4	100%	95%
4	31	9	67%	75%
5	0	10	58%	51%
6	5	8	78%	60%
7	92	6	82%	89%
8	57	8	91%	97%



Neuron

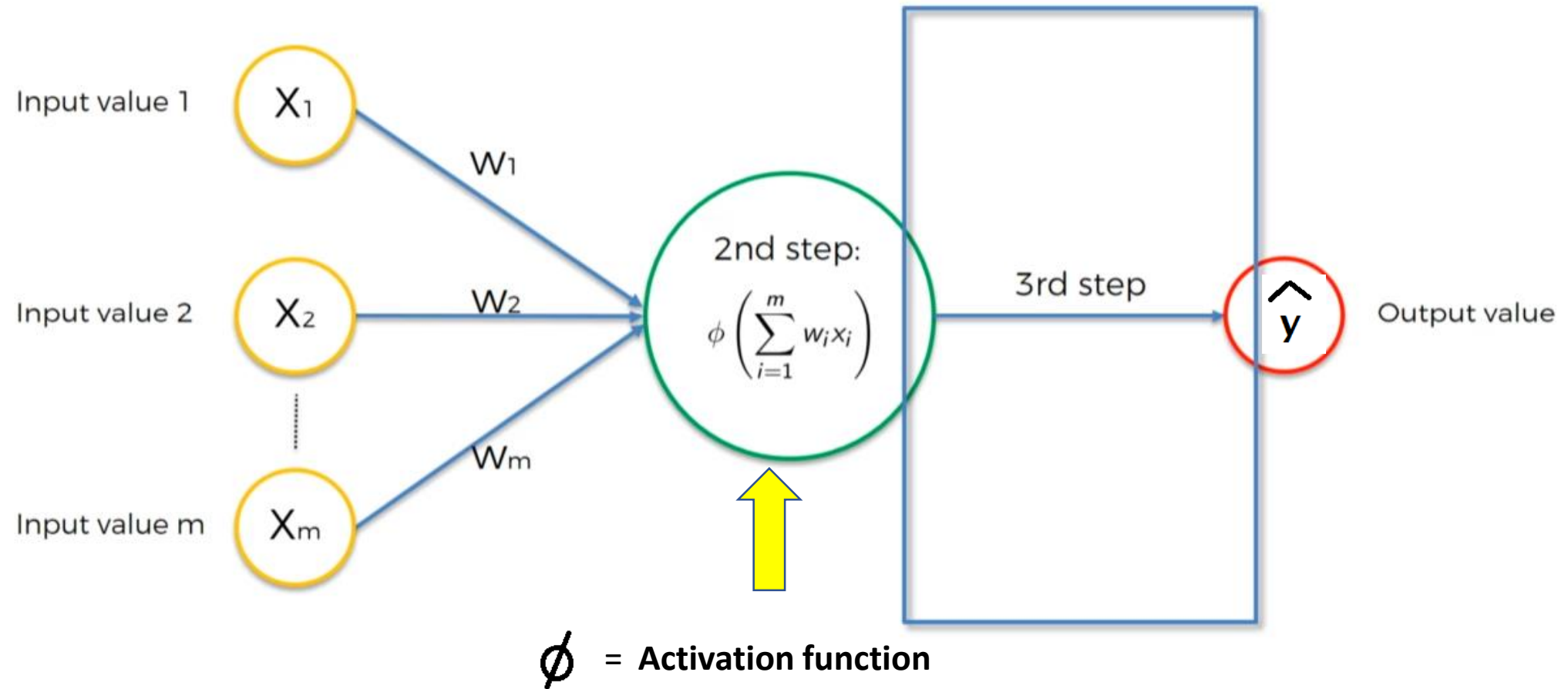


Neuron (Perceptron)



$$\text{Total} = x_1 * w_1 + x_2 * w_2 + x_3 * w_3 + b$$

Activation Function

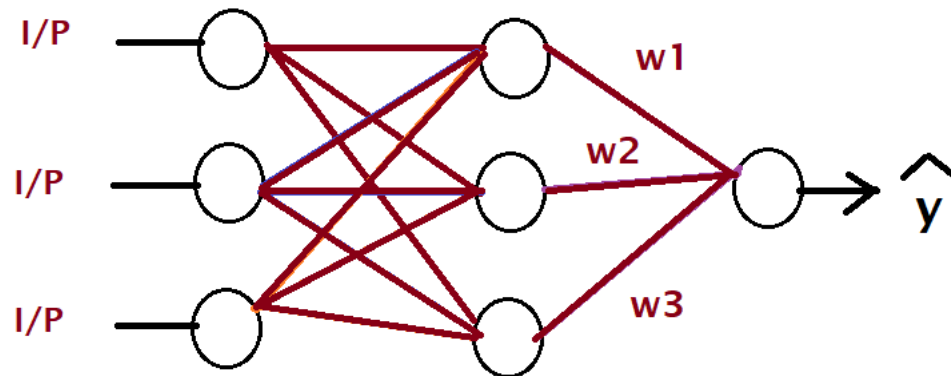


Activation Function

- Their main purpose is to convert a input signal of a node in a A-NN to an output signal
- That output signal can be used as a input in the next layer in the stack
- Specifically in A-NN
 - we do the sum of products of inputs($\mathbf{X}$) and their corresponding Weights($\mathbf{W}$)
 - apply a Activation function $\mathbf{f}(\mathbf{x})$ to it to get the output of that layer
 - feed it as an input to the next layer

① to add some non linearity in output

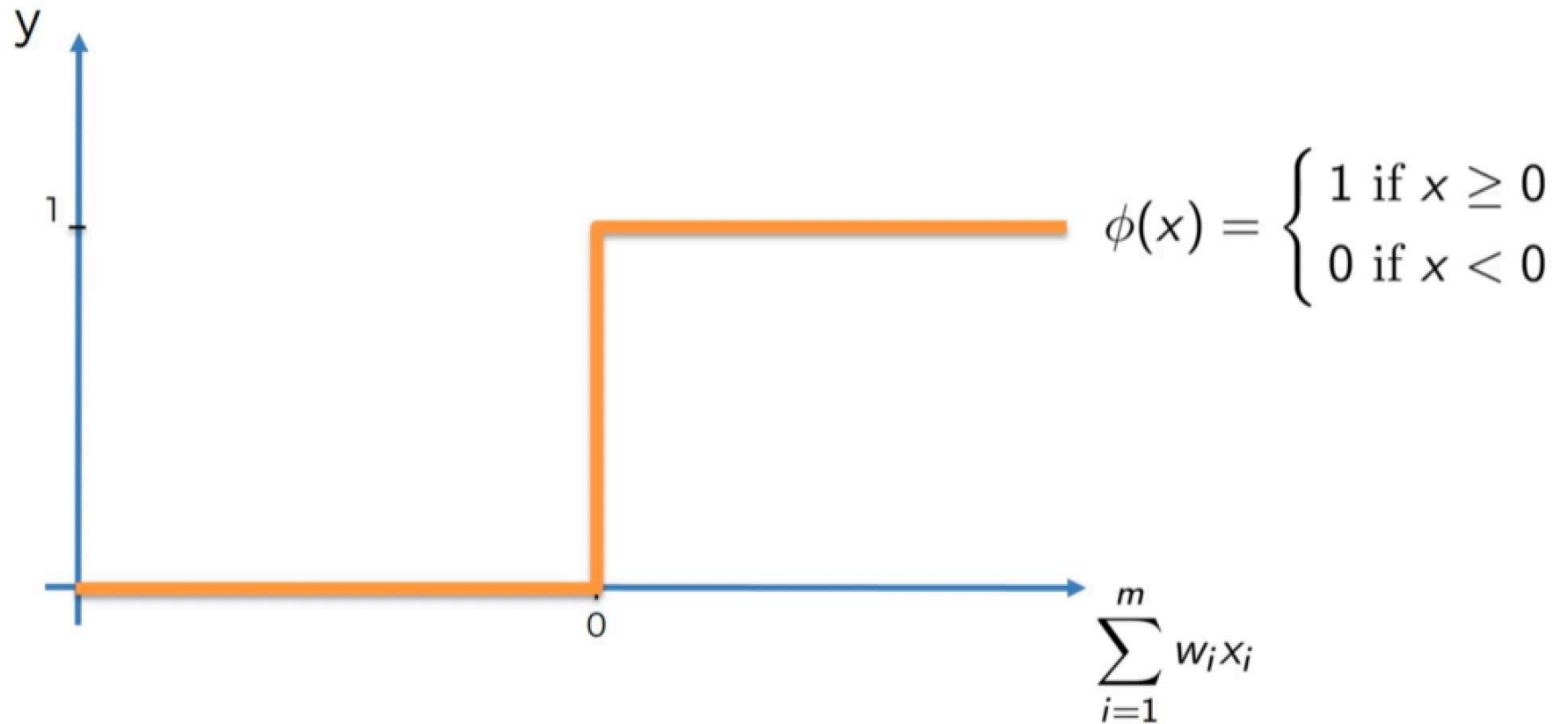
② to restrict o/p into range



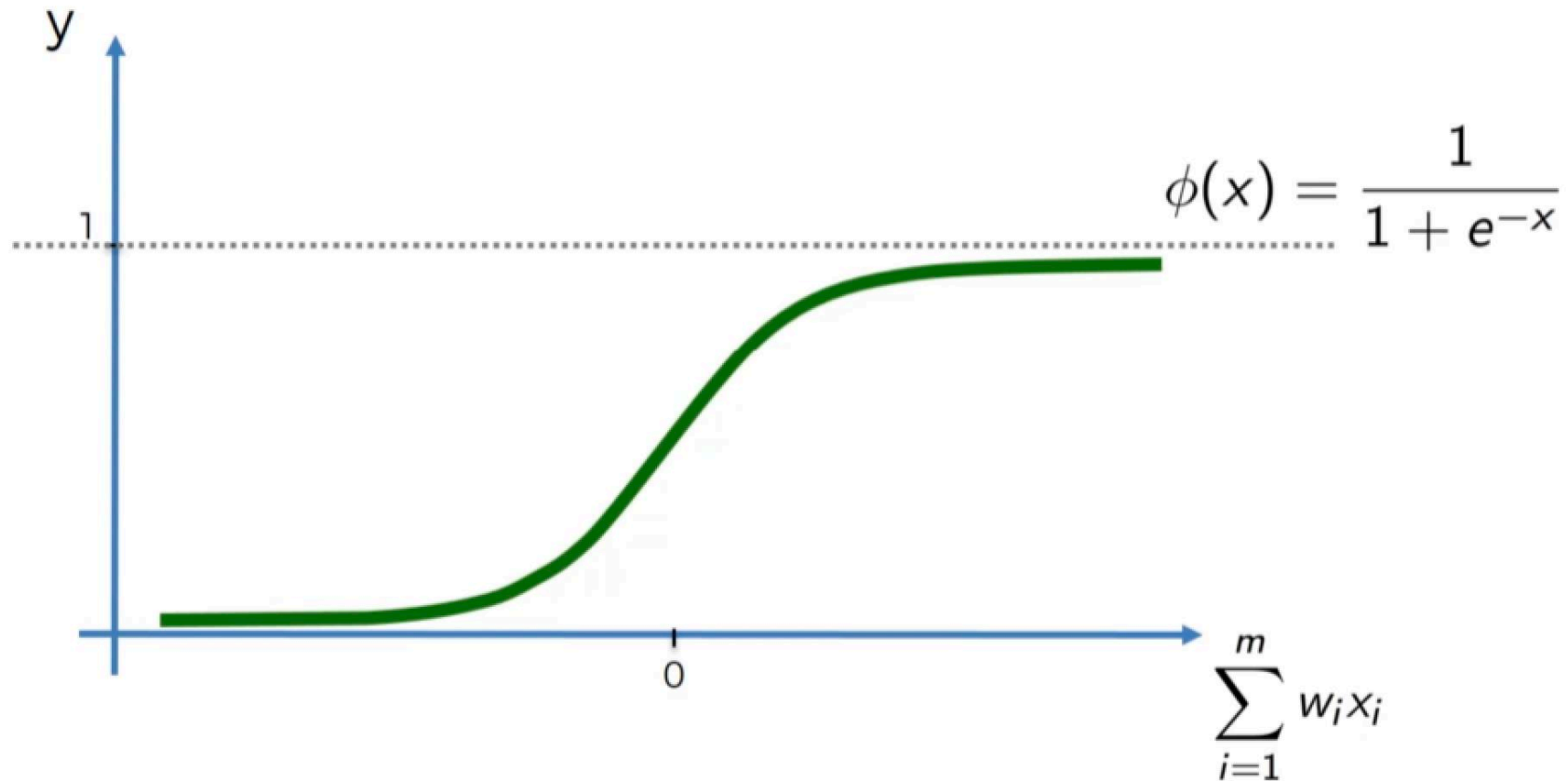
Activation Function



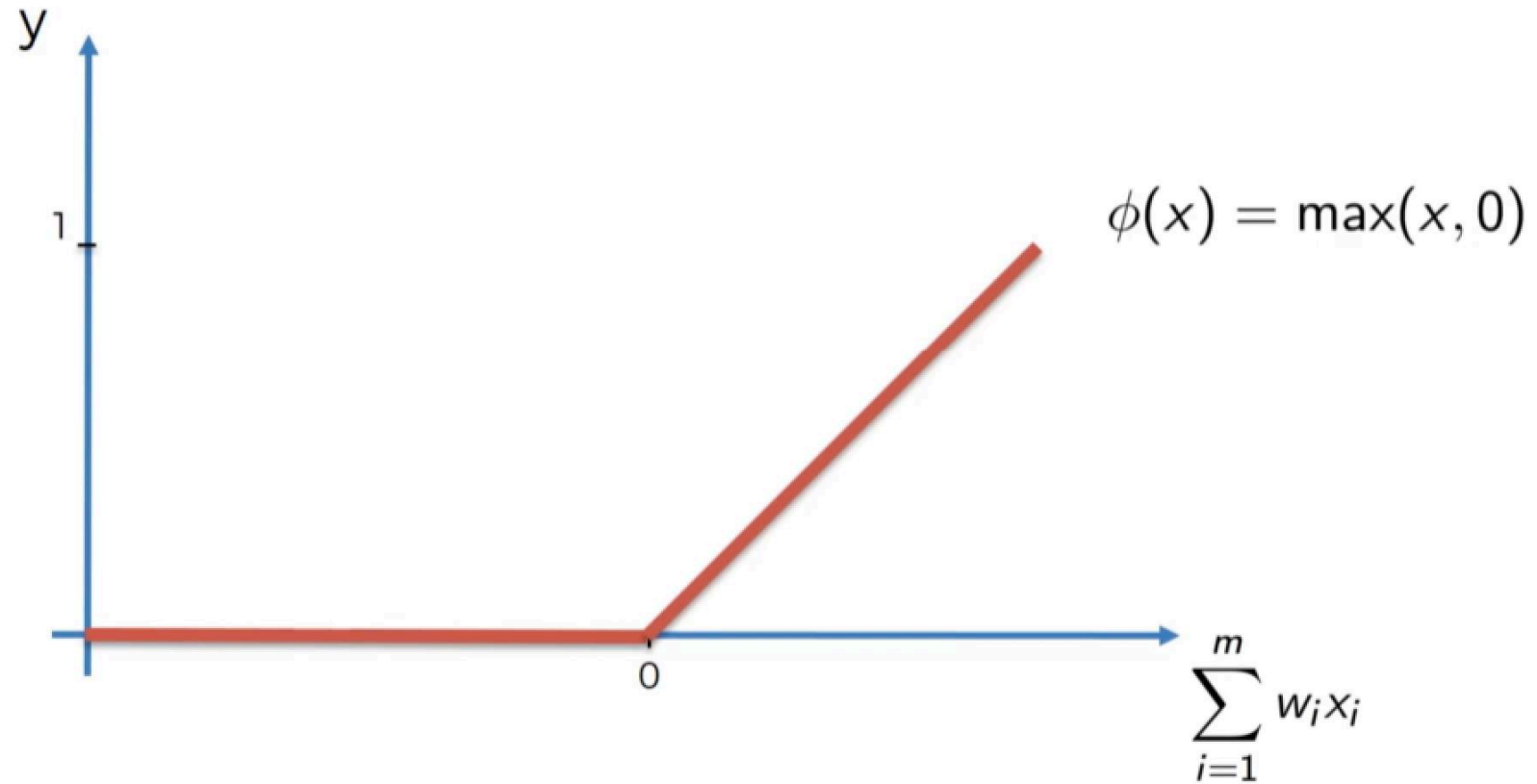
Threshold Function



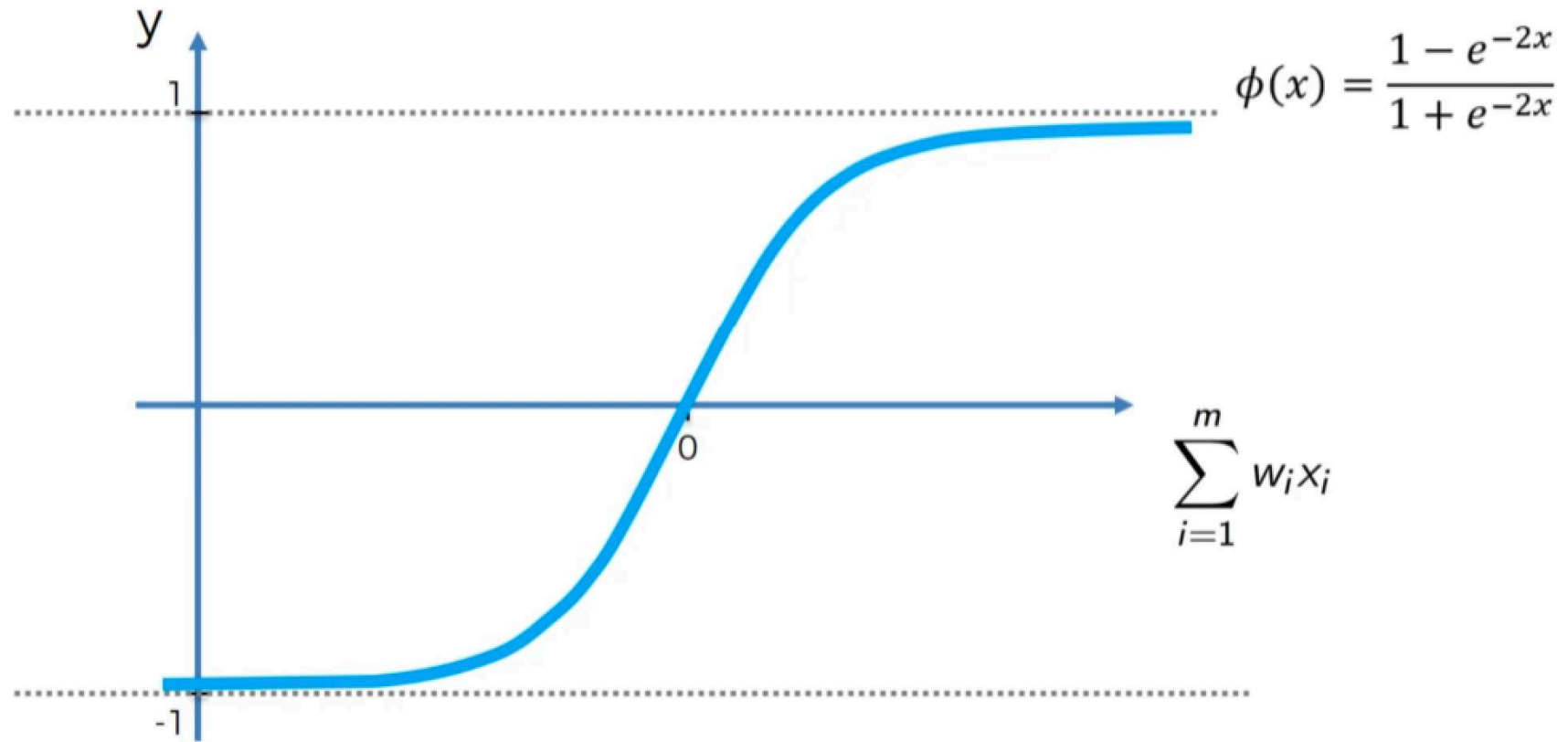
Sigmoid Function



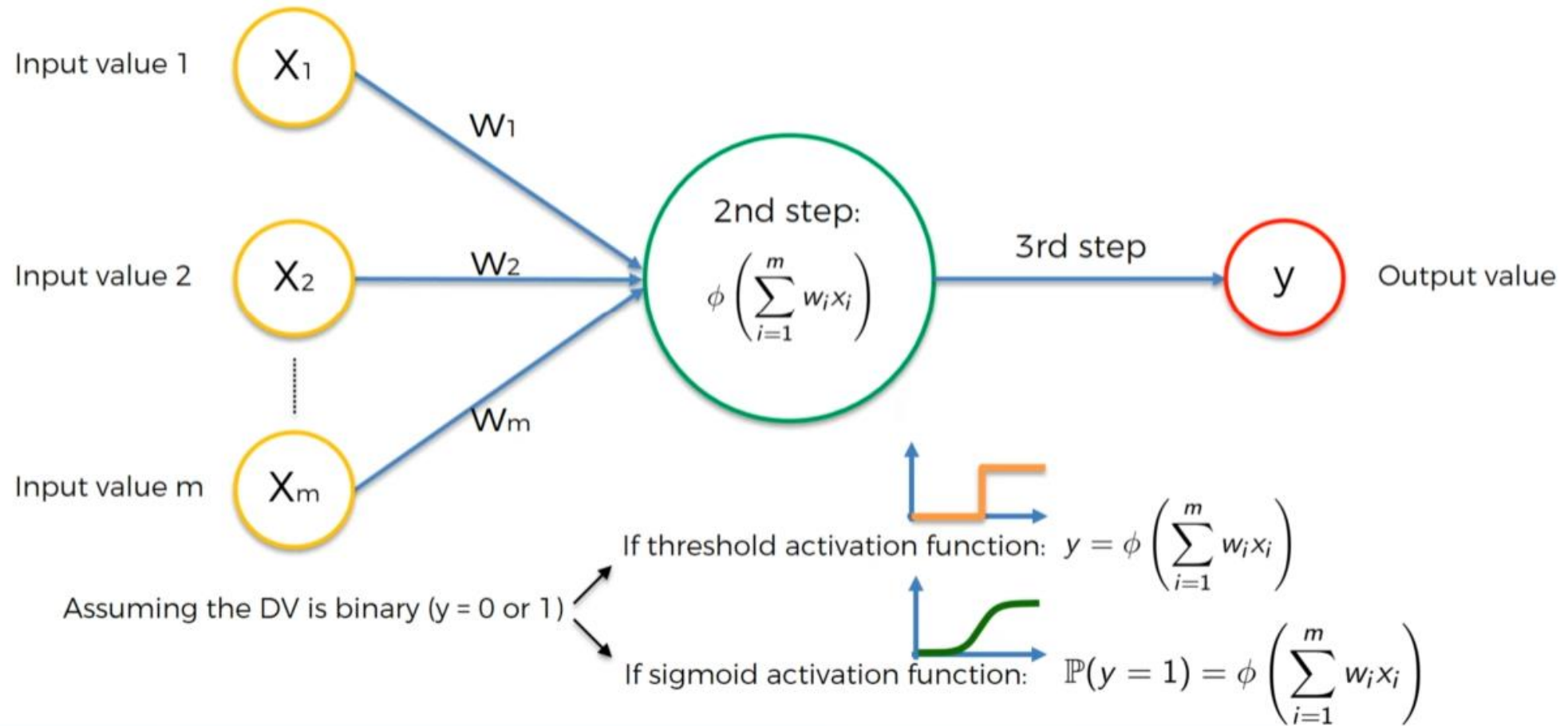
Rectifier Function



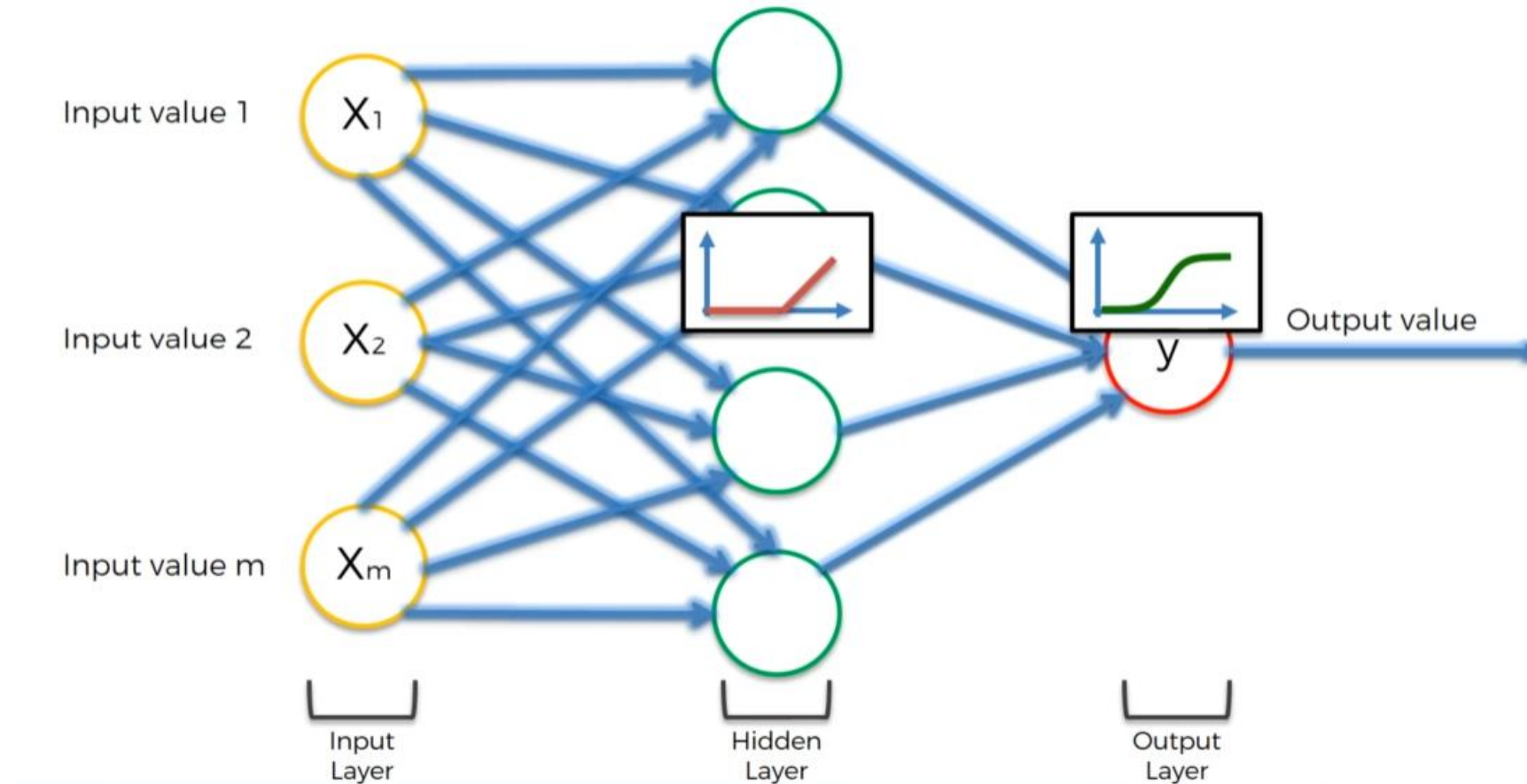
Hyperbolic Tangent Function



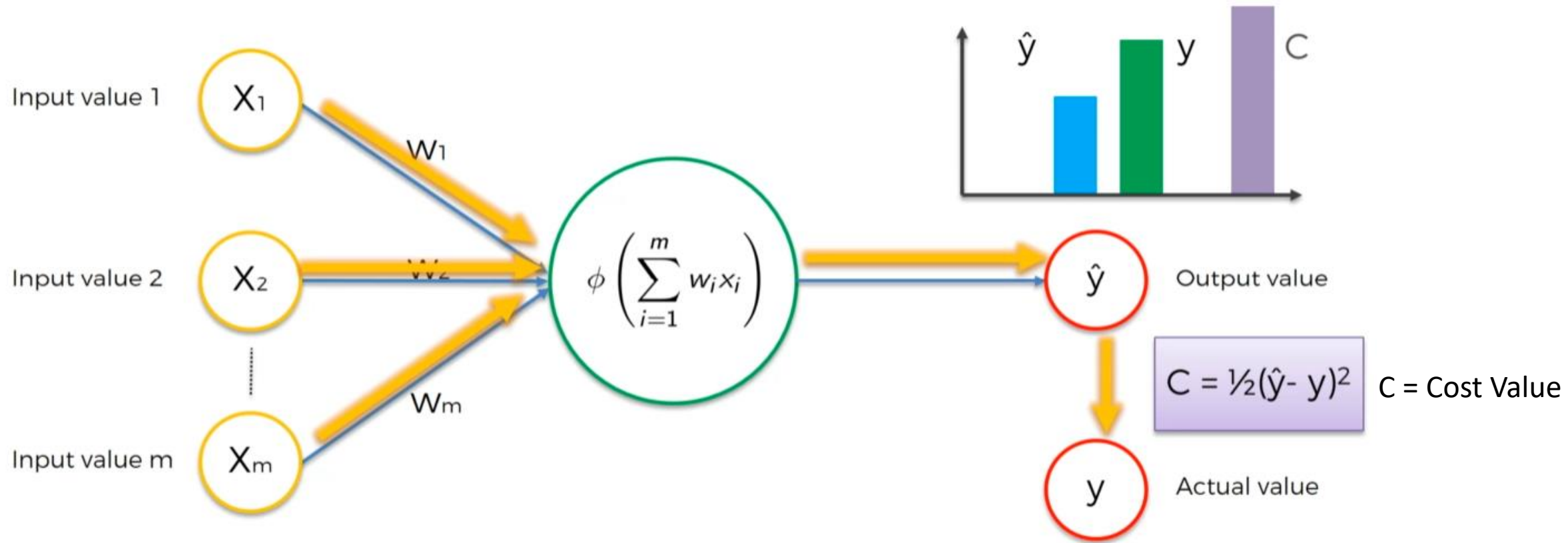
Activation Function



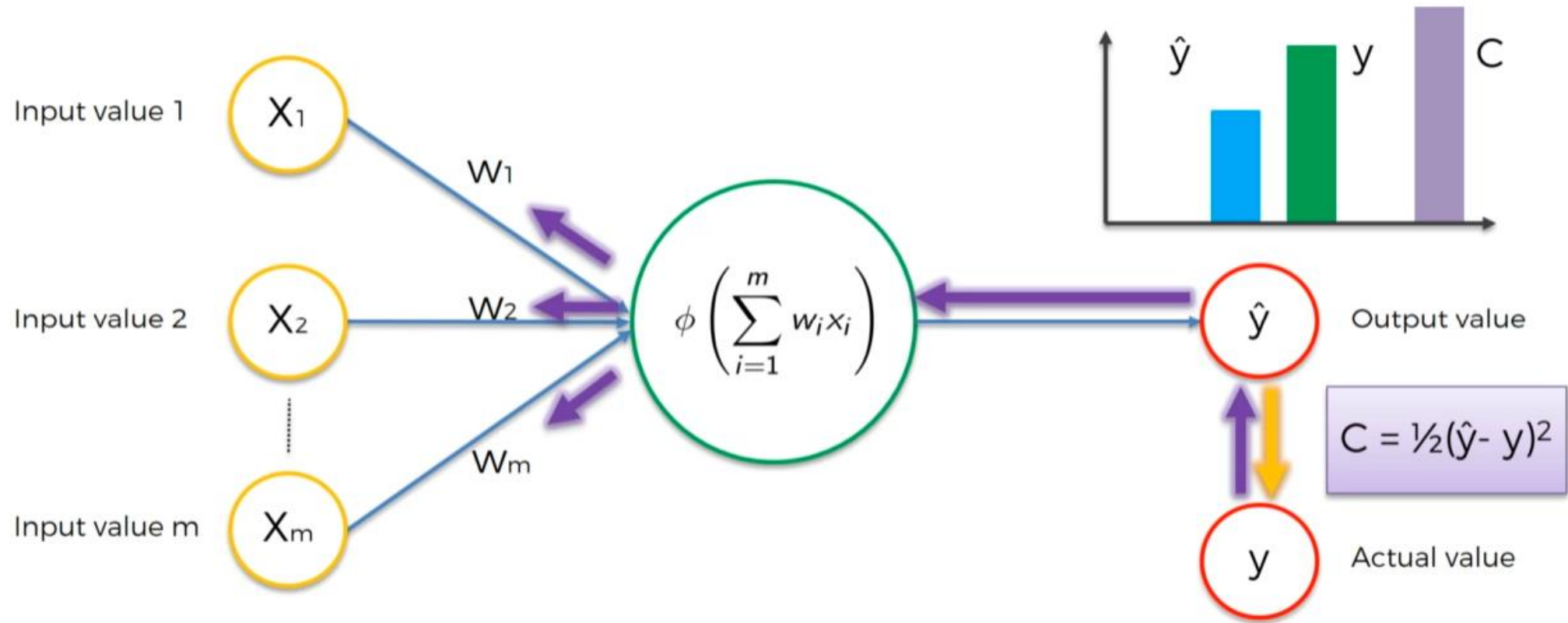
Activation Function



Neural Network - learn

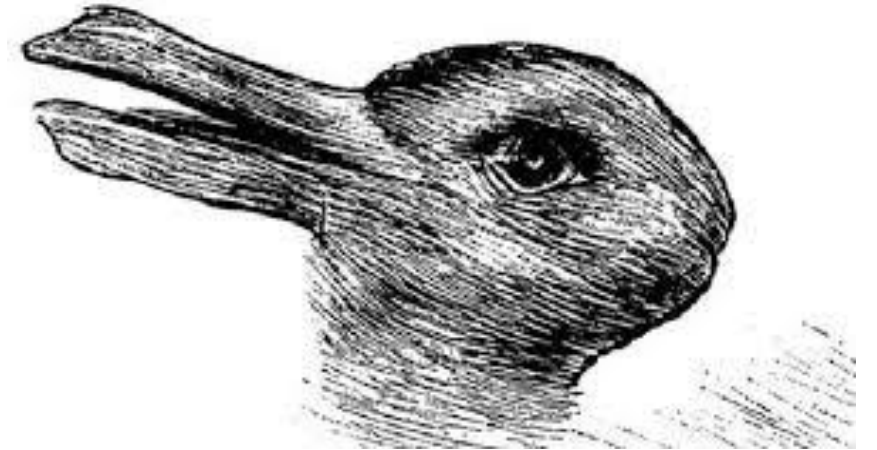
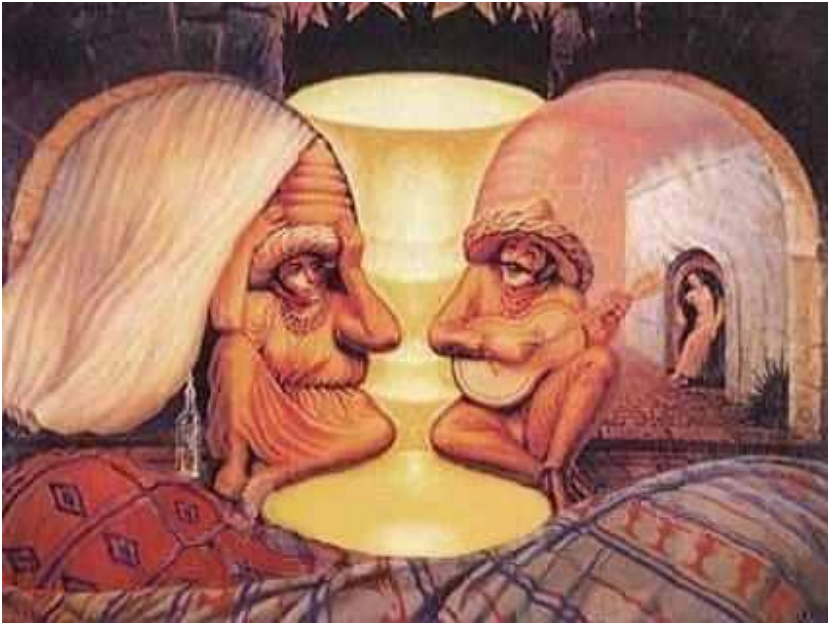


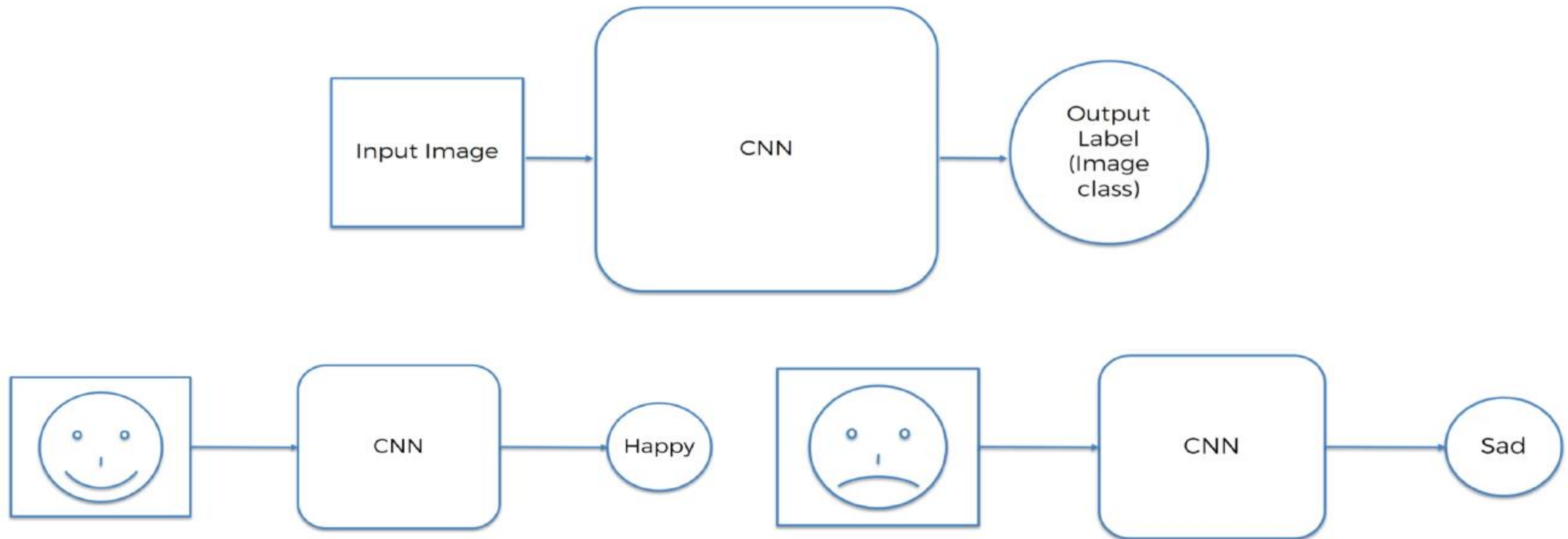
Neural Network - learn



CNN (Convolutional Neural Network)







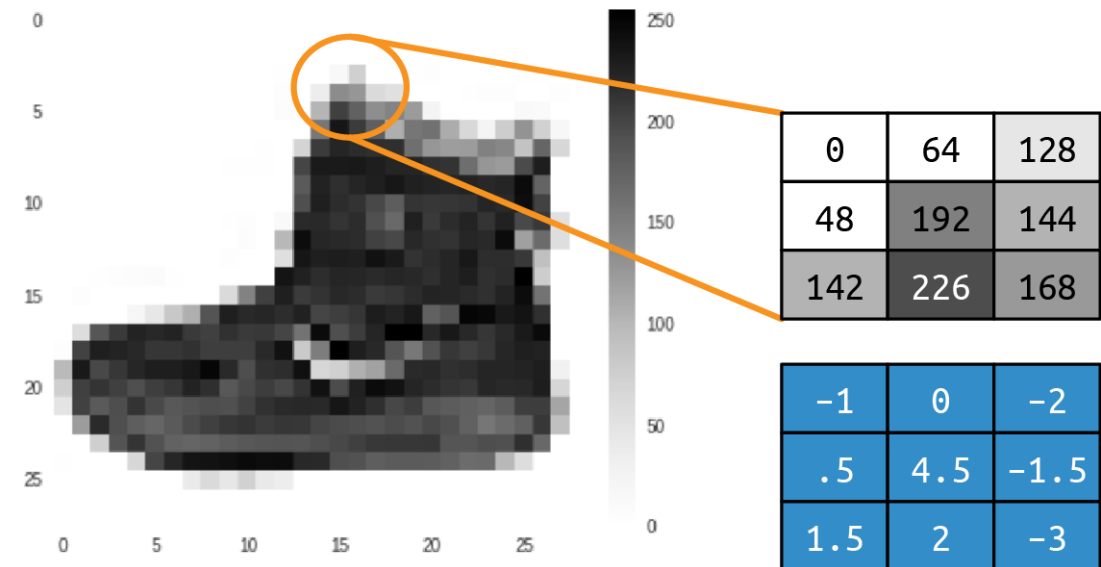
Image

- An image is an artifact that depicts visual perception, such as a photograph or other two-dimensional picture, that resembles a subject—usually a physical object—and thus provides a depiction of it
- It can be seen as a two-dimensional (2D) view of a 3D world
- A digital image is a numeric representation of a 2D image
- It is a finite set of digital values, which are called pixels
- Pixel = picture + element
- SD = 640 * 480
- HD = 1280 * 720 / 1920 * 1080
- UHD = 3840 * 2160
-

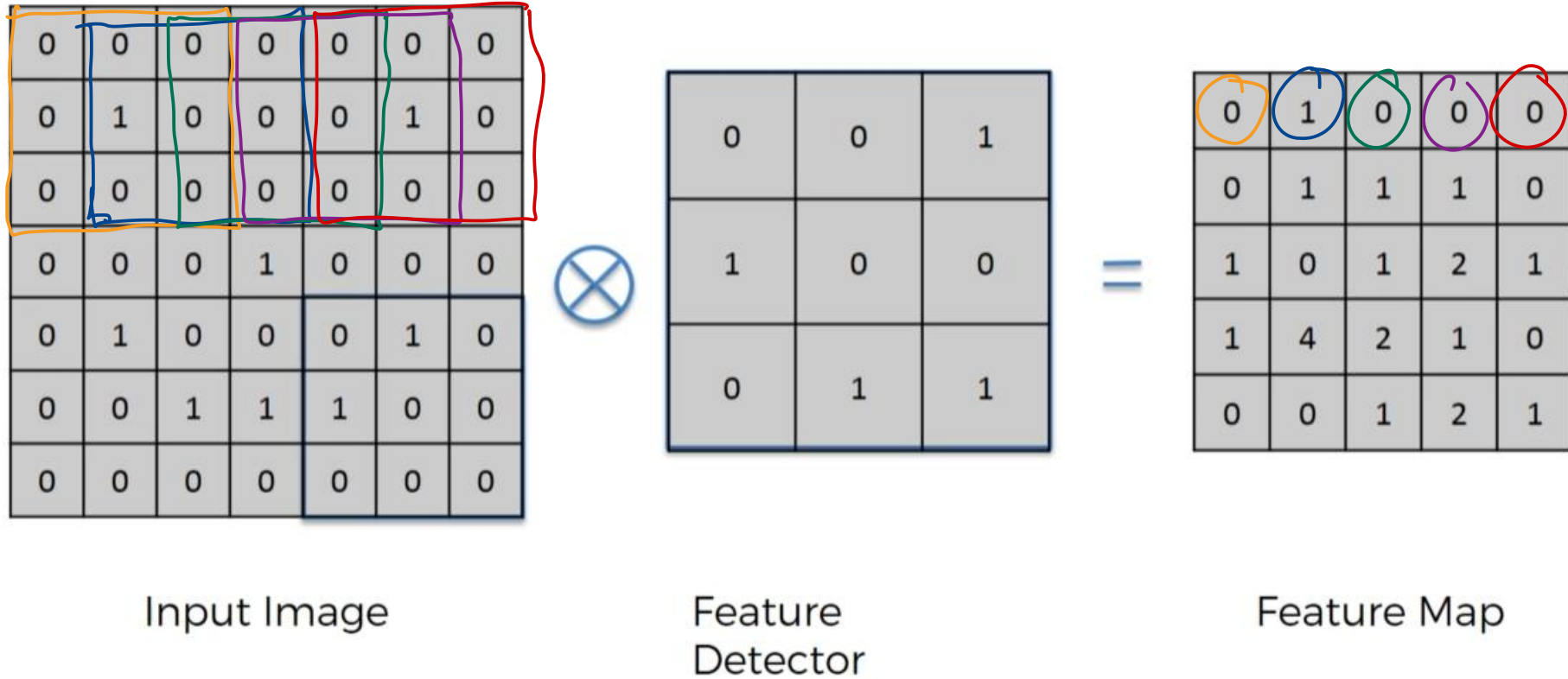


Convolution

- A convolution is simply a filter of weights that are used to multiply a pixel with its neighbours to get a new value for the pixel
- The objective of the Convolution Operation is to extract the high-level features such as edges, from the input image



Convolution



Convolutional Neural Network

- A Convolutional Neural Network (ConvNet / CNN) is a Deep Learning algorithm which can take in an input image, assign importance (learnable weights and biases) to various aspects/objects in the image and be able to differentiate one from the other
- The pre-processing required in a ConvNet is much lower as compared to other classification algorithms. While in primitive methods filters are hand-engineered, with enough training, ConvNets have the ability to learn these filters/characteristics
- The architecture of a ConvNet is analogous to that of the connectivity pattern of Neurons in the Human Brain and was inspired by the organization of the Visual Cortex
- Individual neurons respond to stimuli only in a restricted region of the visual field known as the Receptive Field
- A collection of such fields overlap to cover the entire visual area

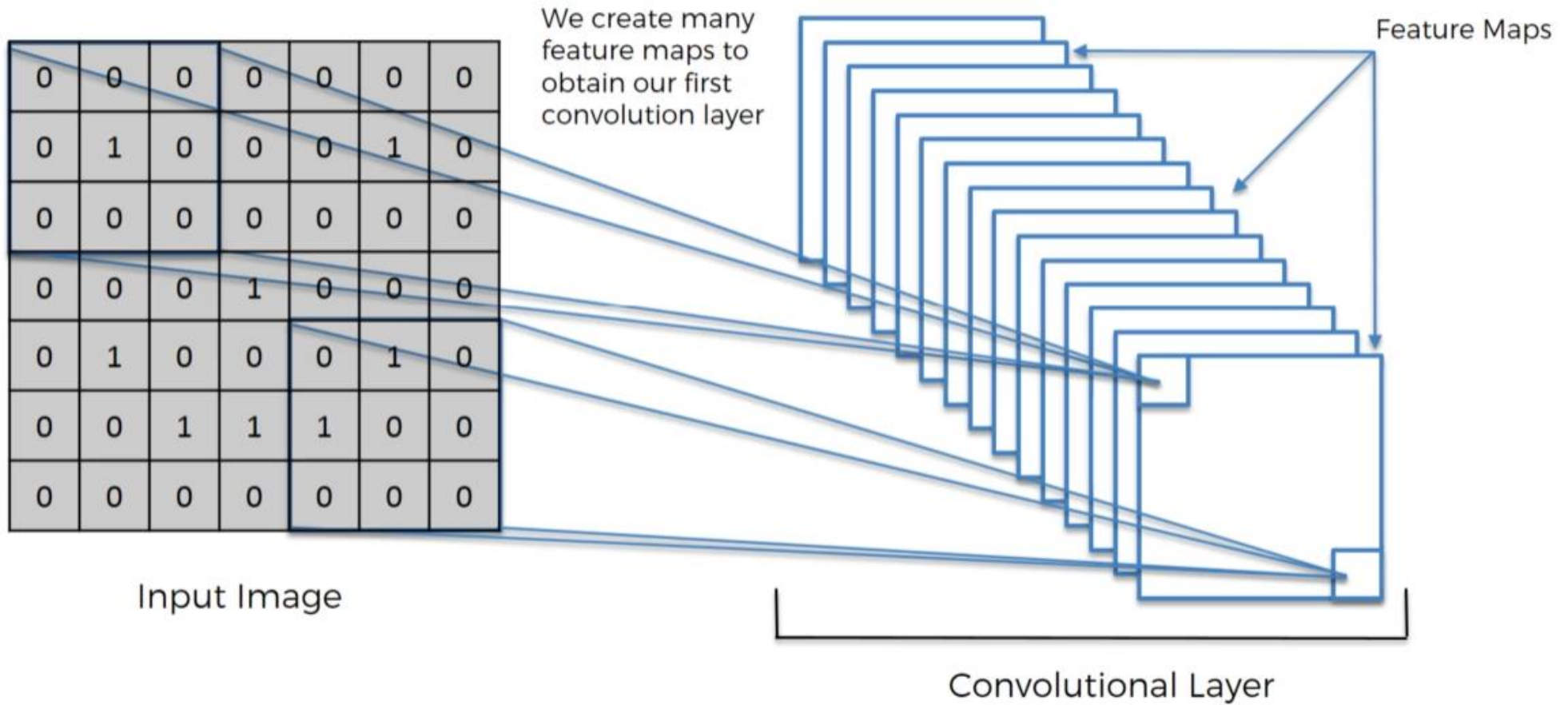


Convolutional Neural Network

- There are various architectures of CNNs available which have been key in building algorithms which power and shall power AI as a whole in the foreseeable future
 - LeNet
 - AlexNet
 - VGGNet
 - GoogLeNet
 - ResNet
 - ZFNet



Convolution Layer

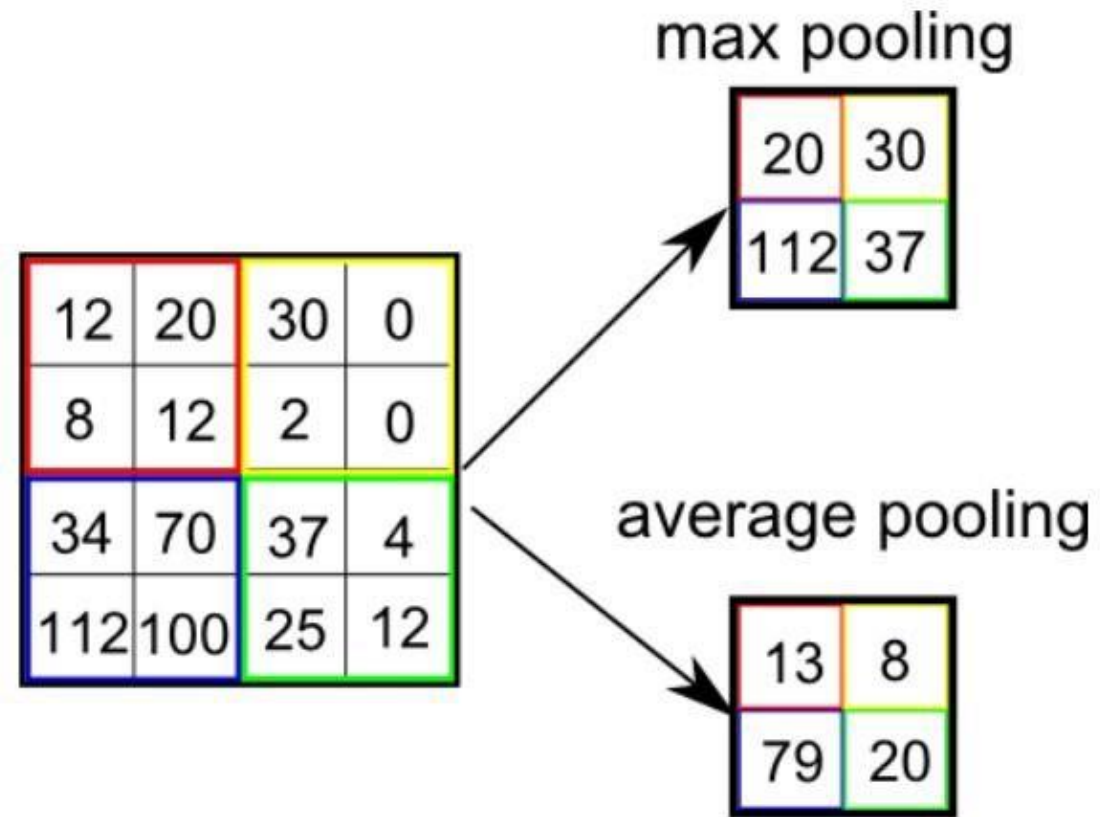


Max Pooling

- Pooling layer is responsible for reducing the spatial size of the Convolved Feature
- Dimension = number of columns
- This is to decrease the computational power required to process the data through dimensionality reduction
- Furthermore, it is useful for extracting dominant features which are rotational and positional invariant, thus maintaining the process of effectively training of the model
- There are two types of Pooling
 - Max Pooling
 - Max Pooling returns the maximum value from the portion of the image covered by the Kernel
 - It also performs as a Noise Suppressant
 - It discards the noisy activations altogether and also performs de-noising along with dimensionality reduction
 - Average Pooling
 - Average Pooling returns the average of all the values from the portion of the image covered by the Kernel
 - It performs dimensionality reduction as a noise suppressing mechanism



Max Pooling



Flattening

1	1	0
4	2	1
0	2	1

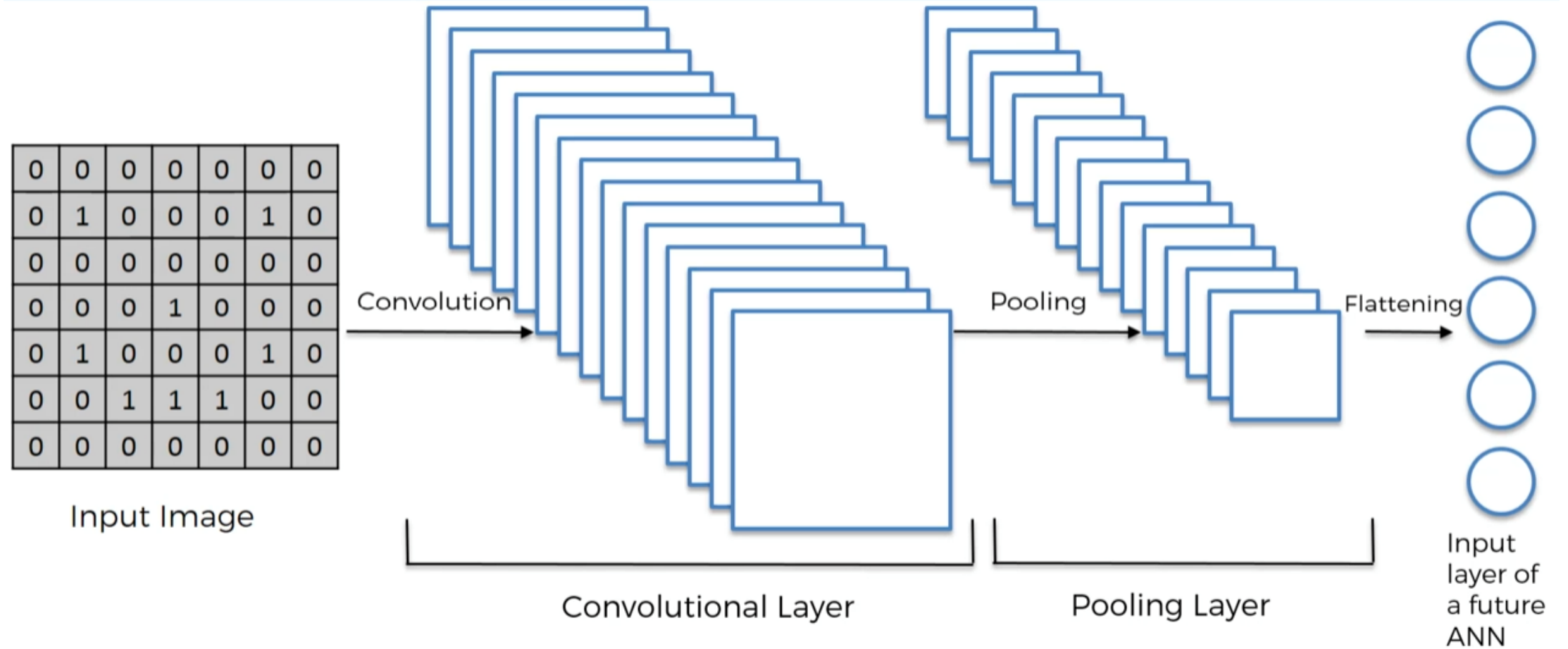
Pooled Feature Map

Flattening

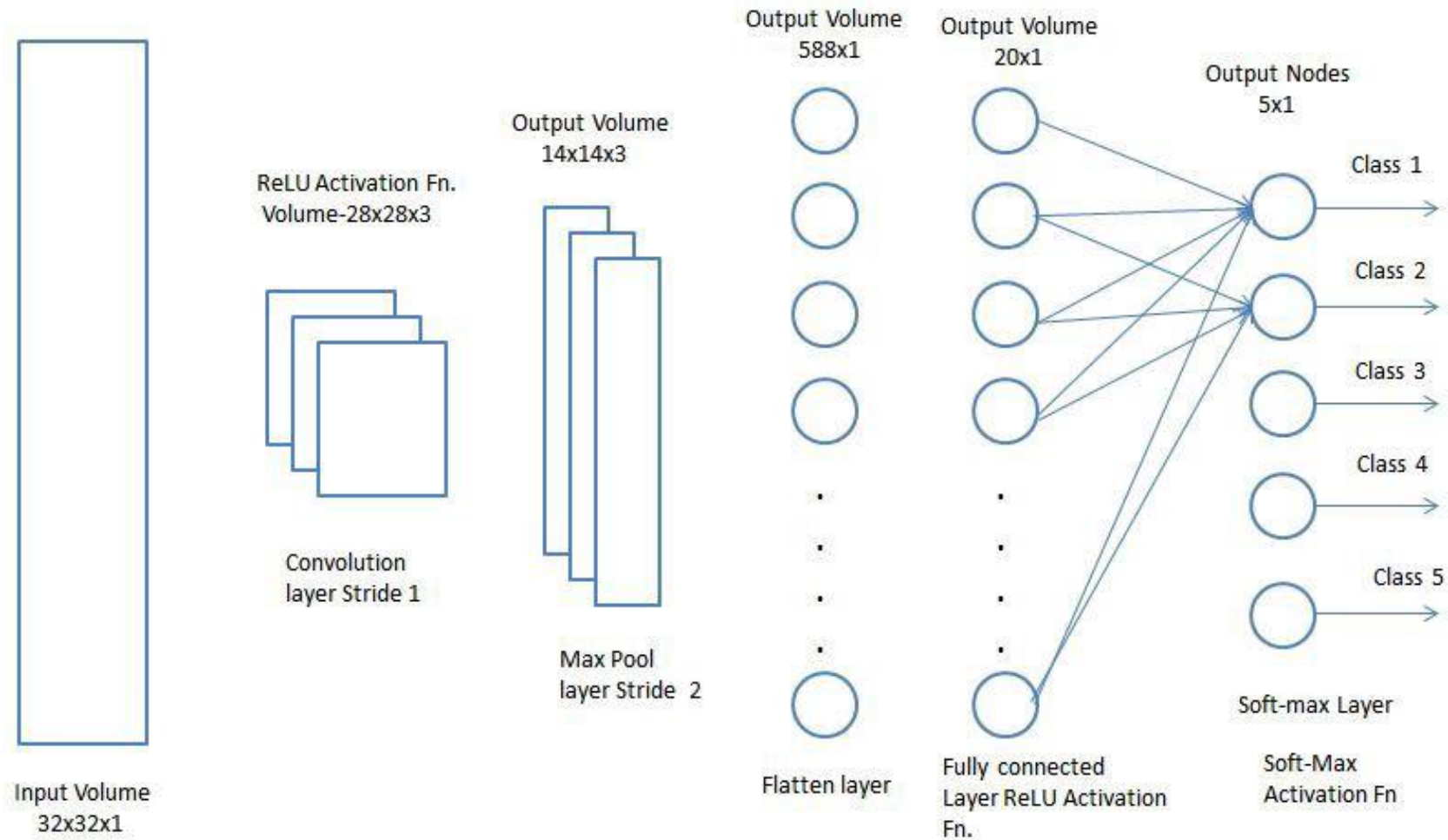


1
1
0
4
2
1
0
2
1

Flattening



Full Connection

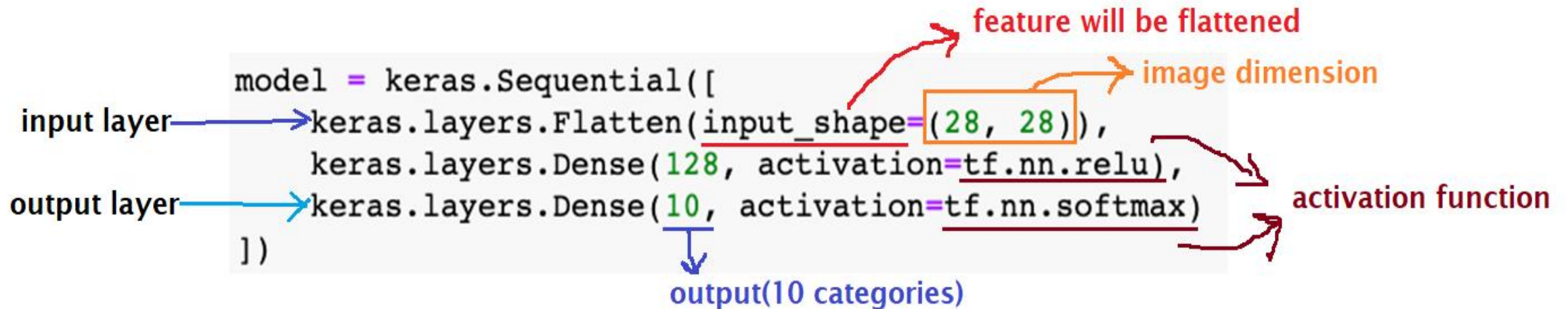


Using TensorFlow



Designing the Neural Network

```
model = keras.Sequential([
    keras.layers.Flatten(input_shape=(28, 28)),
    keras.layers.Dense(128, activation=tf.nn.relu),
    keras.layers.Dense(10, activation=tf.nn.softmax)
])
```



Activation Function

- The **activation function** of a node defines the output of that node given an input or set of inputs.
- **ReLU (Rectified Linear Unit)** : This is most popular activation function which is used in hidden layer of NN
- **Softmax** : Generally, we use the function at last layer of neural network which calculates the probabilities distribution of the event over 'n' different events. The main advantage of the function is able to handle multiple classes.
-



Designing the Neural Network

- The first, Flatten, isn't a layer of neurons, but an input layer specification
- Our inputs are 28×28 images, but we want them to be treated as a series of numeric values
- Flatten takes the image (a 2D array) and turns it into a line (a 1D array)

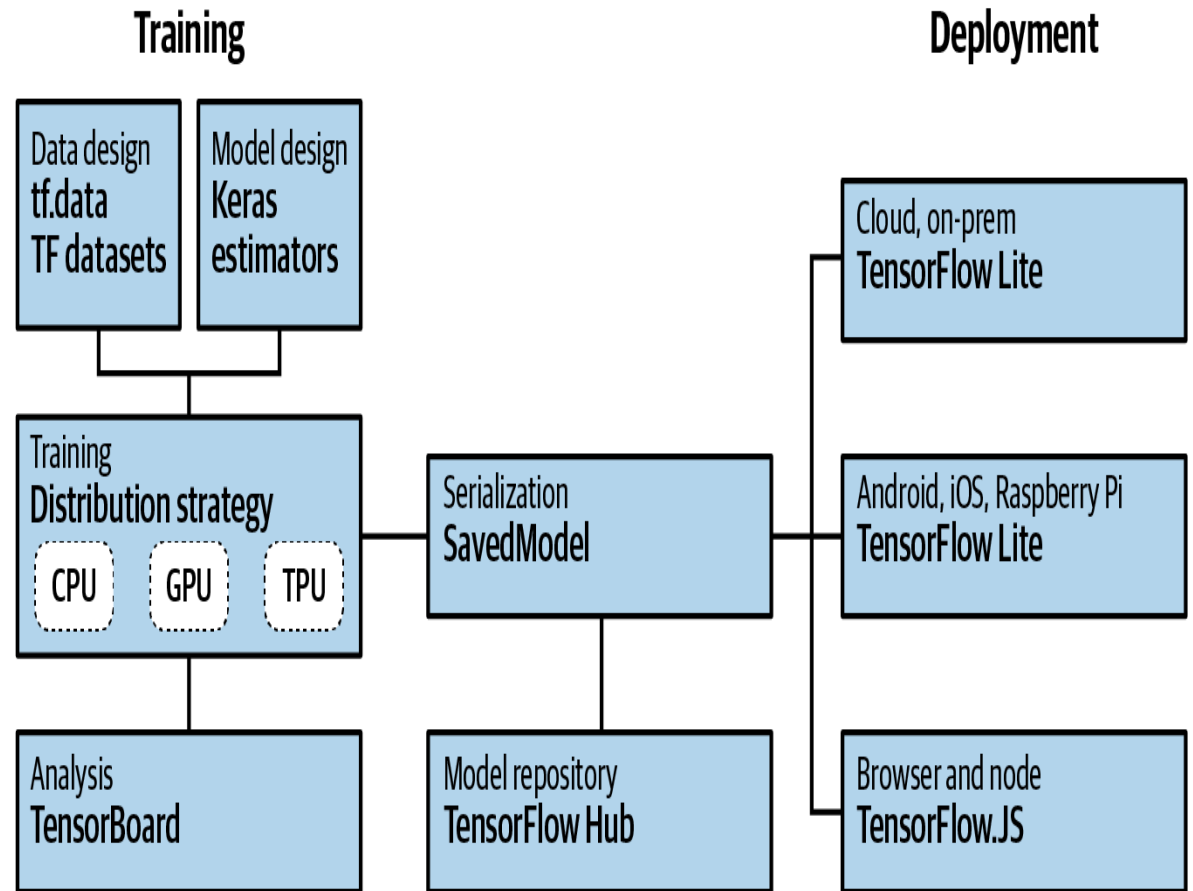
- Finally, there's another Dense layer, which is the output layer which has 10 neurons, because we have 10 classes
- Each of these neurons will end up with a probability that the input pixels match that class, so our job is to determine which one has the highest value

- The next one, Dense (hidden layers), is a layer of neurons, and we're specifying that we want 128 of them
- More neurons means it will run more slowly, as it has to learn more parameters
- It takes some experimentation over time to pick the right values. This process is typically called hyperparameter tuning



TensorFlow

- TensorFlow is an open source platform for creating and using machine learning models backed by Google.
- It implements many of the common algorithms and patterns needed for machine learning, saving you from needing to learn all the underlying math and logic and enabling you to just to focus on your scenario
- It's aimed at everyone from hobbyists, to professional developers, to researchers pushing the boundaries of artificial intelligence
- Importantly, it also supports deployment of models to the web, cloud, mobile, and embedded systems



Saved model format is h5)



First Program using Tensorflow

```
import tensorflow as tf
import numpy as np
from tensorflow.keras import Sequential
from tensorflow.keras.layers import Dense

model = Sequential([Dense(units=1, input_shape=[1])])
model.compile(optimizer='sgd', loss='mean_squared_error')

xs = np.array([-1.0, 0.0, 1.0, 2.0, 3.0, 4.0], dtype=float)
ys = np.array([-3.0, -1.0, 1.0, 3.0, 5.0, 7.0], dtype=float)

model.fit(xs, ys, epochs=500)

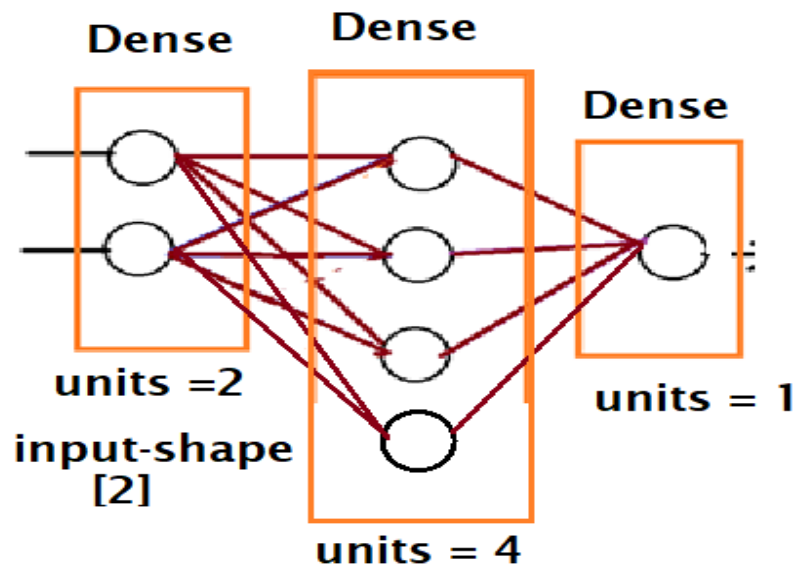
print(model.predict([10.0]))
```

neuron=1

input values are passed as 1D array



Sequential



Sequential = Model

Dense = Layer

Units = neurons

X	-1	0	1	2		10
Y	-3	-1	1	3		?

weights



$$Y = 2 * x - 1$$

$$y = 2 * 10 - 1$$

$$y = 19$$

First Program Explained

- Sequential (Container , ANN)
 - When using TensorFlow, you define your layers using Sequential
 - Inside the Sequential, you then specify what each layer looks like
- Layer
 - A layer is represented as Dense in TensorFlow
 - “Dense” means a set of fully (or densely) connected neurons where every neuron is connected to every neuron in the next layer
- Compiling model
 - The computer starts guessing to create the model (formula) and comparing with the observed value
 - The optimizer used in this stage helps the machine to optimize the guess
 - SGD: stochastic gradient descent, a complex mathematical function that, when given the values, the previous guess, and the results of calculating the errors (or loss) on that guess, can then generate another one
 - ADAM: the optimization is a stochastic gradient descent method that is based on adaptive estimation of first-order and second-order moments





Thank you!!!

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